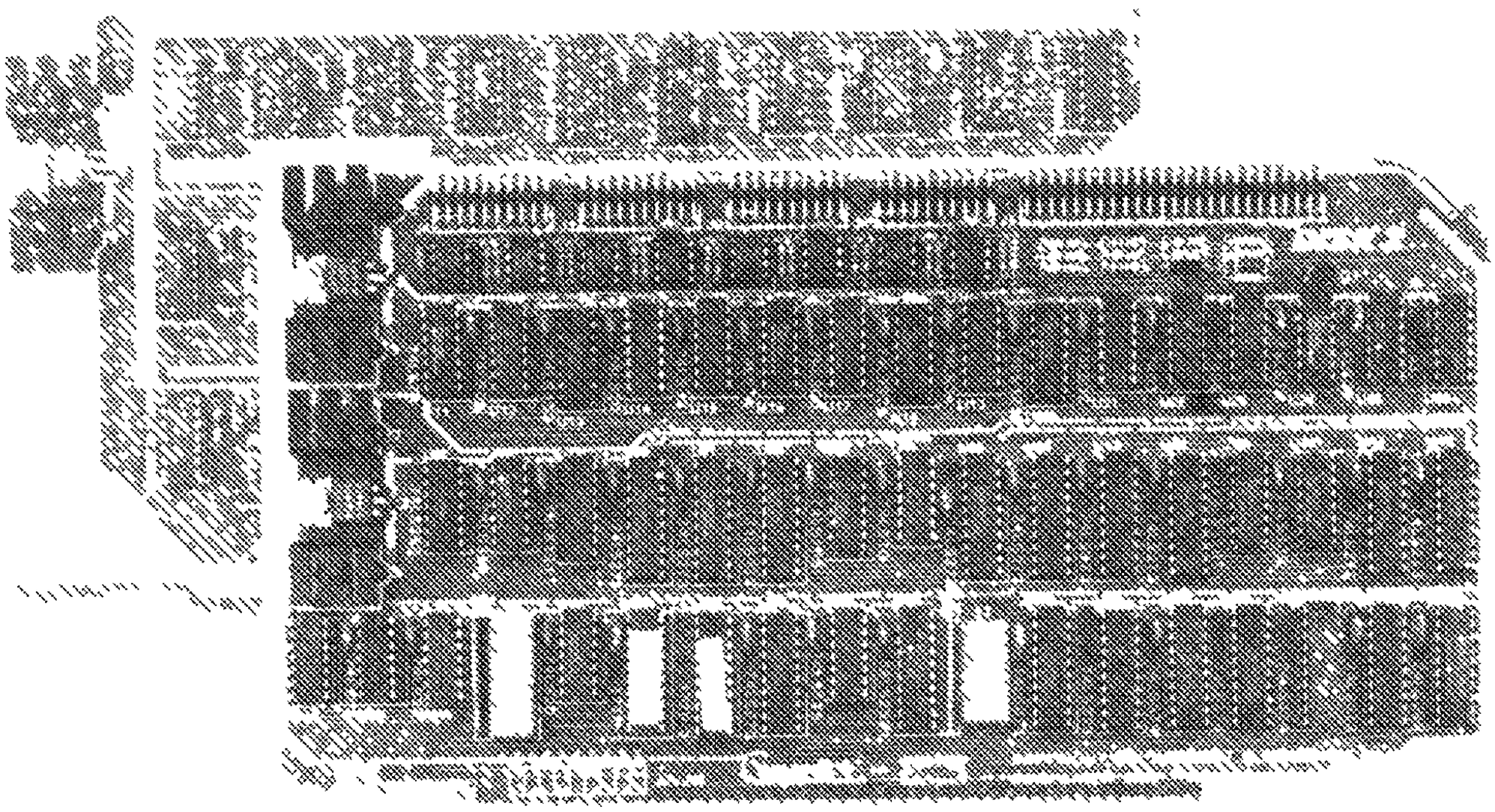
CompuPro

DISK 2™ & SELECTOR CHANNEL™

Technical Manual

IEEE 696/S-100



ARBITRATED 24 BIT DMA HARD DISK CONTROLLER

for Fujitsu 2300 Series Drives

$20.00

A192

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SERVICE AND WARRANTY INFORMATION

DISK 2

ABOUT DISK 2

Congratulations on your purchase of the DISK 2 hard disk controller and the SELECTOR channel. These boards are designed for maximum performance applications featuring full physical and electrical compatibility with the IEEE 696/S-100 specifications. The boards implement a temporary master interface capable of transferring data directly between a hard disk and memory at data rates up to 1 Mbyte per second. The boards may directly access the full 16 Mbyte extended address range of the bus while requiring only three I/O port addresses for their operation.

TECHNICAL OVERVIEW

The DISK 2™ board features a standard IEEE 696/S-100 edge connector along its bottom edge and a set of ribbon cable connectors along the top which will directly accept up to four Winchester type hard disk drives selected from a variety of manufacturers, models and capacities. The DISK 2 contains all of the hardware necessary for control of the drives including the ability to format. The board also contains the necessary hardware to arbitrate for and accept the bus as a temporary master per the IEEE 696/S-100 specification and to transfer data thereon. The SELECTOR channel contains all of the hardware for generation of address, status and control signals required for temporary master operation of the DISK 2. The two board combination functions exactly as a temporary master per the bus specifications. No interconnections (other than the bus itself) are required between the boards for operation and all bus signals are used in their normal way. The DISK 2 performs no useful function without a SELECTOR channel, but the SELECTOR channel may be shared by other controllers designed to use it.

SWITCH SETTINGS AND OPTION SELECTION

	SWITCH 1	FUNCTION	
1	-	Not Used	
2	P3	DMA PRIORITY (MSB)-	
3	P2	DMA PRIORITY	ON = "0"
4	P1	DMA PRIORITY /	OFF = "1"
5	P0	DMA PRIORITY (LSB)-/	
6	20	2048 Byte Sectors--	
7	10	1024 Byte Sectors	
8	5	512 Byte Sectors	ONLY ONE ON
9	2	256 Byte Sectors /	AT ANY TIME
10	1	128 Byte Sectors--/	

	SWITCH 2	FUNCTION	
1	A7	Port Address (MSB)	
2	A6	" "	
3	A5	" "	ON = "0"
4	A4	" "	OFF = "1"
5	A3	" "	
6	A2	" "	
7	A1	Port Address (LSB)	
8	-	Not Used	

PORT SELECTION: The DISK 2 is controlled through two I/O ports. A pair of consecutively addressed ports are selected by setting SW2-1 through SW2-7 according to the seven most significant bits of the desired port address. The CompuPro standard for the DISK 2 is ports C8 and C9.

DMA PRIORITY: When requesting the bus for transfer of data as a temporary master, the DISK 2 will arbitrate with a priority which is set by switches SW1-2 through SW1-5. Priority 15 is the highest while 0 is the lowest. When assigning a priority to the DISK 2 bus transfers, all other DMA devices must be taken into consideration. Then, if the DISK 2 has a higher priority, when any conflict occurs the DISK 2 will win the arbitration. If the other requesting device can complete its operation in only a short time, it could be overall faster to give the slower device a higher priority. Latency time for each DMA device and the amount of bus access time required by each DMA device will help determine which device should have which priority for optimum system throughput.

SECTOR SIZE: The DISK 2 may read, write and format sector sizes from 128 bytes to 2048 bytes. The desired sector size is selected by SW1 positions 6-10 according to the table. Only one of these switches may be on at a time. (Changing sector sizes also requires adjustments in the drives and software, and reformatting the drives).

In addition to setting switches as described above, a number of options are selected by means of jumpers.

INTERRUPT VECTOR SELECTION: Use of interrupts is not required by the CP/M supplied with the DISK 2/SELECTOR CHANNEL board set. But, if the use of an interrupt is desired to allow the CPU to process other tasks concurrent with a DISK 2 operation, installation of a jumper in one of the locations from J7 to J14 will result in ATTN, (the "done" flag) asserting one of the vectored interrupt lines VI0 through VI7 respectively. The flag may be reset through the control register (See Programming the Disk 2). If the use of an interrupt is desired, and a vectored interrupt controller is not available, jumper J15 will result in ATTN asserting INT. An interrupt on this line may be sensed directly by the CPU. Note that ATTN will come up asserted after a RESET.

WRITE CLOCK PHASE: All of the Shugart compatible drives (such as the Memorex 100 series and the Fujitsu 2300 series) require jumper J6 (between U25 & U26) in the C* position while Shugart SA4000 series drives require that J6 be in the C position.

FORMATTING A HARD DISK

Before using your hard disk, you must format it with the "DISK2" utility provided with your CP/M 80 and CP/M 86 system disks. The "DISK2" program will format, verify and test your hard disk drive. The different options that can be used are in the following list.

USAGE: DISK2 {options}

At least one option must be specified such as the drive type. You can mix any of the options.

Options consist of:

m10	Set drive type to Fujitsu 10 Mbyte
m20	Set drive type to Fujitsu 20 Mbyte
m26	Set drive type to Shugart 26 Mbyte**
m10m	Set drive type to Memorex 10 Mbyte
drive #	Format selected drive
format	Format headers
data	Write out data fields with E5H
test	Perform a data field test
seek	Perform a seek test
all	Perform format, data test, seek test
skew #	Set skewing of disk to specified number This option goes along with hardware settings.

*** Although the DISK 2 program will format these drives, The CompuPro CP/M 80 and CP/M 86 systems are not configured for them.**

If an option is not specified then the following defaults are used:

```
Defaults -   M20
             skew of 2
             sector size 1024
             drive 0
             no tests or formatting

Examples -   disk2 format data
             disk2  m20 all
             disk2  data
             disk2  format data drive 1 skew 3 m10
```

JUMPER & SWITCH SETTINGS

Check to see if your Memorex 100 has the Fujitsu 2300 LSI or MSI board revision.

LSI FUJITSU 2300 & NEW MEMOREX 100 SERIES

S1	SW1		SW2		SW3		SW4		SW5	
(Jumper)	OFF	ON	OFF	ON	OFF	ON	OFF	ON	OFF	ON
2 to 3	< 1		< 1			1 >	< 1			1 >
	2	>	< 2			2 >	< 2			2 >
	< 3			3 >	< 3		< 3			3 >
	< 4			4 >	< 4		< 4		< 4	
	< 5		< 5		< 5		< 5			
	< 6			6 >	< 6		< 6			
	7	>	< 7		< 7		< 7			
	< 8		< 8			8 >	< 8			

MSI FUJITSU 2300

S1	SW1		SW2		SW3		SW4	
(Jumper)	OFF	ON	OFF	ON	OFF	ON	OFF	ON
2 to 3	< 1		< 1			1 >	< 1	
	2	>	< 2			2 >	< 2	
S2	< 3			3 >	< 3		< 3	
(Jumper)	< 4			4 >	< 4		< 4	
1 to 2	< 5		< 5		< 5		< 5	
	< 6			6 >	< 6		< 6	
S3	7	>	< 7		< 7		< 7	
(Jumper)	< 8		< 8			8 >	< 8	
1 to 2								

SHUGART SA4000 JUMPER

LSB	1	2	3	4	5	6	7	8	9	10	11	12	MSB
	out	out	in	in	in	out	out	in	in	in	out	in	

X	DS1/2,3,4	ST	RY	IX	T	C
out	in/out	in	in	in	in	out

SC / BC	D	E	S1/S2	4H	E1,E2,E3	R
in / out	in	out	in/out	shunt out	no conn.	in

S	F	SL	GR	DL1
out	out	in	in	out

PROGRAMMING THE DISK 2

All functions of the DISK 2 are controlled through a pair of I/O ports whose addresses are selected by SWITCH 2. The lower addressed port is the control/status port. Data written to this port is placed in the CTL register while reading this port will return the contents of the STATUS register. The upper port is the DATA port. Writing to this port will send a byte of data to a data register previously selected by setting the CTL register. Reading the DATA port will cause a STEP pulse to be sent to the previously selected drive (drive selection will be discussed below). No valid data is returned by the read.

CONTROL/STATUS REGISTER

BIT	CTL (WRITE)	STATUS (READ)
7	ATTN*	ATTN*
6	RUN	TIME OUT
5	OP2	CRC ERROR
4	OP1	OVER RUN
3	OP0	READY*
2	FAULT CLR	SEEK COMPLETE*
1	US1	WRITE FAULT*
0	US0	TRACK 0*

* indicates an active low signal

While the STATUS register may be read at any time, it might be in either a transparent or a latched state. If it is transparent, the data returned is the current state of the register's inputs. If it is latched, data returned was the input data at the time the register was latched. The current state is indicated by the ATTN* bit, ATTN* true (low) indicating a latched condition. The ATTN* bit may be controlled through the CTL register to make the STATUS register transparent.

There are four registers which may be written through the DATA port. To write a specific register, the register number is first stored in the CTL register bits OP1 and OP0 as shown in the table below. This will direct a byte of data written on a subsequent transfer via the DATA port to the appropriate register.

OP1	OP0	DATA REGISTER	
0	0	DRIVE	SEE STATUS/CONTROL
0	1	CYLINDER	CHART ABOVE FOR
1	0	HEAD	BIT ASSIGNMENT
1	1	SECTOR	

DRIVE SELECTION AND STATUS: Up to four drives may share the control cable which is connected to the 50 pin ribbon connector and runs "daisy chained" to each of the drives. There is a unique DRIVE SELECT line in the cable for each of the drives. Only one of these select lines may be active at a time. The selected drive will respond to the control output lines on the cable and drive its status onto the input lines. The select lines are driven from the four high order bits of the DRIVE register. Bit 4 selects the first drive, bit 5 the second etc. The READY* bit in the status register is driven by the READY line of the selected drive. Similarly the status lines WRITE FAULT*, SEEK COMPLETE* and TRACK 0* all reflect the status of the selected drive.

HEAD POSITIONING: Most drives require that the controller software remember the current track. To position the read/write heads at a new track, the drive is selected and the DATA port is read a number of times equal to the difference between the current track and the new track. Each read from the DATA port will send a STEP pulse to the selected drive causing its heads to move one track. The direction of the STEP is controlled by the OP2 bit of the CTL register. OP2 in the CTL register drives the DIRECTION* line in the control cable. Most drives use stepping motors or other positioning devices with limited step rates. (See drive specifications for maximum step rates).

The starred (*) paragraphs below are features that may be available on your drives and are for use by the more experienced system programmer.

* Many of the newer drives feature a fast seek mode. These drives will accept and accumulate STEP pulses at a very high rate without moving the access mechanism. After all of the pulses are received, the SEEK is performed at a much higher average track per second rate than would be achieved stepping one track at a time. (See drive specifications.) The DISK 2 board can accommodate this feature if your drive has it.

HEAD SETTLING TIME: After a seek, the hard disk requires a significant amount of time (in the order of one revolution) for the heads to settle before reading or writing is attempted. The software must provide this delay starting with the receipt of SEEK COMPLETE. This time period can be found in your drive specifications.

* Some of the newer drives provide a jumper option which will delay the SEEK COMPLETE indication by the required head settling period eliminating this burden from the drive software.

* The DISK 2 automatic functions are interlocked to SEEK COMPLETE. As an alternative to the use of a timer for generation of the head setting delay in a multiprogramming application, the disk software may issue a NULL command immediately following the step pulses of the seek. The NULL command will not begin until the drive indicates a SEEK COMPLETE. NULL has no function other than to fail with a TIME OUT indication after two index pulses from the drive. This completion would interrupt the CPU indicating that the drive was ready to accept a read or write command. If the available drives include the option of delaying SEEK COMPLETE by the head settling time, the disk software may issue a READ DATA or WRITE DATA immediately following the step pulses. Again, the interlock will hold off the automatic operation until SEEK COMPLETE is indicated.

READ SELECTION: Prior to any data transfer operation to the disk, the software must select a read/write head. The drives will have at least one and sometimes two heads for each surface of each platter used for data storage. The control cable includes 4 head address lines. The lines are driven by the lower four bits of the DRIVE register, bit 0 being the least significant. The required nibble selecting one of up to sixteen heads, as appropriate for the drive, will normally be stored in the DRIVE register at the same time that the drive is selected since this is the function of the upper nibble in the register.

AUTOMATIC OPERATIONS: On the disk, each data field is preceded by a unique header field which was written when the disk was formatted. A read or write operation consists of searching the entire track of the currently selected drive and head for the header record of the desired sector. Once the header is found, the data field immediately following the header may be read or written. Because of the high data transfer rate of the disk, all of these operations occur too quickly for direct software control. For this reason, the DISK 2 contains an FSM (finite state machine) which performs these functions automatically.

THE NULL COMMAND: (COMMAND 00) This command was mentioned above under head settling time however its primary application is to align sectors relative to index during formatting. It is issued in the same way as a READ DATA except with a NULL operation code. The CYLINDER, HEAD and SECTOR register are not required and no SELECTOR CHANNEL programming is required because no data is transferred. A ready drive must be selected as a source for index or the DISK 2 will fail to complete the operation.

THE READ DATA COMMAND: (COMMAND 01) Prior to issuing a READ DATA command, the location of the required sector must be determined and the drive, track and head selection processes completed so that once the read begins, the sector may be found. These selection processes have all been described above. Since the READ DATA will be performed automatically, the memory address to which the data will be transferred and the identity of the sector header must all be determined before the operation is begun. (See the SELECTOR CHANNEL manual for programming considerations). The header fields contain a three byte code. During a search, these bytes are compared against the contents of the CYLINDER, HEAD and SECTOR registers. These are DATA registers whose contents may be loaded as described above. The contents of the CYLINDER, HEAD and SECTOR registers have no affect on the selection process, they are only used as a source of data for matching headers.

Once all of the data has been loaded, the command byte may be written to the CTL register. The command byte must contain the following:

US1 & US0:	These bits must contain the drive number and be consistent with the contents of the DRIVE register.
FAULT CLR:	Must be 0.
OP2 - OP0:	Must contain the operation code for the desired function. (See Op Code Table).
RUN:	Must be 1 indicating an automatic operation.
ATTN*:	Must be a 1 (false) making the status register transparent and indicating busy.

Once the command has been issued, the CPU is free to perform other tasks. The DISK 2 will transfer the sector to memory automatically once it is found. Completion of the operation is indicated by the ATTN* bit of the STATUS register which may optionally result in an interrupt.

RESULT STATUS: It is possible that an automatic operation may not complete satisfactorily. After any automatic operation, the STATUS register should be read, and it contents which were latched at the moment of completion should be evaluated. The resulting status from a successful read will be a 3 (or a 2 if track 0 is being read). If an error has occurred, the bits from the STATUS register should be evaluated with the following priority:

READY* (Bit 3) If the drive had lost its ready state, no data transfers may occur. TIME OUT will probably also be true. Check for wrong drive selected. May also be caused by a power line dip or drive failure.

TIME OUT (Bit 6) Automatic commands will be terminated and the time out indication set if two index pulses are received from the drive before an operation is completed. This allows between one and two complete passes of the track to find a header. TIME OUT would indicate that the header was not found. Since soft errors may occur, the original command should be retried a few times. If the TIME OUT persists, a READ HEADER command may be used to read any header on the track. This header found would indicate if the wrong track or head was being searched. A wrong track could result from a drive seek error. If the track was wrong, the drive should be homed to track 0 and the seek performed again. If all of the above are correct and the header is still unreadable, a hard error is indicated.

OVER RUN (Bit 4) This bit indicates that the request for use of the bus was not acknowledged quickly enough and some or all of the data may have failed to make it to memory. If the OVER RUN was due to collisions with other DMA devices or CPU response to the bus request, it may be resolved through retries. If this status persists and none of the data was transferred, the SELECTOR CHANNEL programming and bus DMA capability should be verified. If some of the data is being transferred, possibly incorrect, the combination of the bus clock rate and memory speed as controlled by wait states may not be fast enough for the drive being used.

CRC ERROR (Bit 5) This bit indicates that the CRC check word computed on the data as it was read from the disk did not match the check word which followed the data field. This again may result from a "soft" read error. The operation should be retried.

THE WRITE DATA COMMAND: (COMMAND 02) The WRITE DATA procedure is the same as the READ DATA procedure except the command sent to the CTL register must have the write operation code in the OP2 - OP0 bits. An additional status may occur on a write operation, namely WRITE FAULT. This status signal is bit 2 in the STATUS register. This status may result from attempting some kind of illegal operation but would normally indicate a drive malfunction. WRITE FAULT must be reset through FAULT CLEAR in the CTL register before the disk will respond to any new commands. After each write operation the status should be checked and if there are any errors they should be checked in the same priority as during a read operation with the WRITE FAULT error checked just after the READY* bit.

THE WRITE HEADER COMMAND: (COMMAND 03) This is the primitive command used in formatting a drive. The command will write a sector header record upon receipt of the next sector pulse from the disk. The three data bytes for the field will be read directly from memory. To format a track, the program must issue a sequence of WRITE HEADER commands, one for each sector on the track. Each command must be issued immediately after the preceding one completes so that it will be pending when the next sector pulse is received from the disk. As an aid to issuing the commands quickly, the data field for all of the headers on a track should be computed in advance and placed in a buffer. As each command is executed, it advances the address register in the SELECTOR CHANNEL by three bytes leaving it pointed at the data for the next field. If tracks are to be formatted with sectors in a specific position relative to index, the first WRITE HEADER command should be issued immediately upon completion of a NULL command (which occurs at index). While the data placed in the header fields is arbitrary as far as the hardware is concerned, the cylinder, head and sector numbers are recommended. (See track format and state timing.)

THE READ HEADER COMMAND: (COMMAND 04) As described above in resolving a TIME OUT status, one may issue a READ HEADER command. This command will transfer the three data bytes of the next header field which passes under the head directly to memory. The requirements for issuing the command are the same as for a READ DATA except the CYLINDER, HEAD and SECTOR registers need not be loaded and only three bytes will be transferred. As in a READ DATA COMMAND, error checking should be done after each operation.

THEORY OF OPERATION

ON THE BUS SIDE - THE I/O PORTS

The S-100 bus data strobes PWR* and PDBIN are ORed to produce a common strobe. This strobe is further qualified by the octal comparitor U51 if the bus address lines A1-A7 match the port address selected by SW2, and the bus status is I/O. The decoder U40 distributes the combined board select and strobe signal into two write strobes and two read strobes using the least significant bus address line A0 and the I/O status.

Data from the CPU is buffered onto the board by U36 which is a latch, but always in its transparent mode. This internal data bus goes to all of the software setable registers, namely CTL (U35), CYLINDER (U55), HEAD (U54), SECTOR (U56) and DRIVE (U38). The CTL register is strobed directly by one of the two write strobes derived above while the data registers CYLINDER, HEAD, SECTOR and DRIVE receive their strobes from a second decoder, also in U40, which distributes the second write strobe according to the two least significant bits of the operation code.

The first of the two read strobes enables the STATUS register (U37) to drive its contents directly onto the CPU's data input bus. The second "read" strobe does not do a read at all. It drives the STEP* line in the disk interface cable through buffer U50.

ON THE BUS SIDE - THE DMA INTERFACE

The DISK 2 performs all data transfers between the disk and memory directly as opposed to techniques in which the CPU would read or write a byte of data at a time from the controller and then write or read the byte as appropriate to main memory. While CPU data transfer worked perfectly well for floppy disk transfer rates, the byte rate of a hard disk does not allow nearly enough time for the bus cycles required to do the two data transfers and fetch the required instructions. One alternative to the speed problem is to read the data to a RAM buffer on the controller and then, after the disk transfer has completed, the CPU could transfer data between the buffer and memory. While this technique is sometimes used, most applications requiring a high performance disk are better served by eliminating the need for the CPU to transfer the data at all, not just postponing it.

DMA operations are described in the IEEE 696/S-100 specifications as temporary master operations because the controller will temporarily take over as bus master generating all bus control signals normally provided by the CPU and transferring data to memory on the CPU's data output bus or receiving them from memory on the data input bus. The DISK 2 performs these operations in conjunction with the SELECTOR CHANNEL. While the SELECTOR CHANNEL is physically a second board, the two boards are required to work together to provide the temporary master as described in the specifications.

When the finite state machine anticipates a requirement to access memory, it will assert the signal K4. The gate U34 will wait for the bus hold and hold acknowledge signals HOLD* and HLDA to both be logically false (indicating that no other temporary master is currently using the bus) and then set APRIO, the assert priority latch, starting the bus arbitration process. APRIO asserts the bus HOLD* line and enables the system of gates consisting of U43, U44 and U45 to drive the DMA priority set by switches SW2 positions 2-5 onto the DMA arbitration lines DMA0* - DMA3*. The function of the gates is to disable less significant bits of the priority if another device is asserting a higher priority. If any of the bits are disabled, the IMHI (I'm high) will be false. By the time HLDA is received from the CPU, IMHI will have settled. If the arbitration was lost (another device of higher priority wanted the bus at the same time) U15 will clear APRIO. If this occurs, APRIO will set again for another try as soon as the bus is available.

Meanwhile, over in the SC (SELECTOR CHANNEL), a circuit monitoring the arbitration and comparing the winning number with the contents of its priority register will observe the DISK 2 winning the bus, and perform the required exchange of bus mastership with the CPU. The DISK 2, on finding that the SC had disabled the CPU status lines will set a latch (U18) which in turn will enable the DISK 2's driver for the bus control line pSYNC. This completes the process of obtaining the bus. Nothing else will happen until it is time to start a cycle.

When the finite state machine wishes to transfer a byte, it will set the DRQ flip-flop (U16). U17 will allow the first stage of U20 to clear at the next rising edge of the bus clock. The output of this stage will enable pSYNC via U20. At the following bus clock, the same output will cause the next stage of U20 to clear. The output of this stage will negate pSYNC also via U20. In response to the pSYNC, the SC will generate the strobes required for the transfer, extending them as required to satisfy any wait requests. U20 will latch the occurrence of the strobe in the third stage of U20 and use the fact that this stage is set and the absence of strobes to clear DRQ with U17. This process of setting DRQ, generating a pSYNC and clearing DRQ will repeat for each byte to be transferred between the DISK 2 and memory.

When the finite state machine determines that it will no longer require access to the bus, it will negate Q4. This will clear APRIO disabling the assertion of HOLD* and the priority lines. This is the signal for the SC to tranfers the mastership of the bus back to the CPU. The latch U18 will assure that the DISK 2 will drive the pSYNC line until the SC releases the status disable line SDSB*.

ON THE DISK SIDE - THE DISK INTERFACE

The DISK 2 has five connectors along its upper edge for the connection of drives. The four twenty pin connectors J1 - J4 carry high speed differential data and clock signals between the DISK 2 and the corresponding drive. J5 carries single ended open collector control and status lines between the controller and all of the connected drives.

Four of the signals in this connector, DRIVE-SEL1* through DRIVE-SEL4* enable one of the connected drives to use the balance of the lines which are multiplexed. The select lines are driven from the upper nibble of the DRIVE register (U38) via the buffers U22 and U23. The lower nibble of the DRIVE register drives HEAD-SEL1* through HEAD-SEL4* also using buffers in U22 and U23. In a given application, less than four drives may be connected and the drives may have fewer than sixteen heads. In addition, the STEP* control line which was pulsed by reading the DISK 2 data port as described above, the controller drives four other control lines. DIRECTION* which controls the direction during stepping and FAULT-CLR* which is used to reset a drive WRITE-FAULT status are driven from the OP2 and FAULT-CLR bits in the CTL register through buffers U22 and U23. The last two control output lines READ-GATE* and WRITE-GATE* are generated by the finite state machine (to be described below) and share the buffers in U22 and U23. Five input lines from the control cable are terminated and used on the DISK 2. They are DRIVE-READY*, TRACK0*, WRITE-FAULT*, INDEX* and SECTOR-CLK*.

Each of the signal connectors on J1-J4 have a pair of differential line drivers and a pair of differential line receivers associated with it. The line drivers send the differential signals WRITE-CLOCK and WRITE-DATA to the disk. Each differential signal has a + and a - phased signal line associated with it. The WRITE-DATA and WRITE-CLOCK for all of the drives are driven from the common internal single ended signals WRITE-DATA and SEL-CLK. The line receivers receive the differential signals READ-DATA and PLO-CLK from each of the drives. The READ-DATA signals are converted to single ended signals RD1* - RD4* by the line receivers. RD1* - RD4* are then ORed by U14 to provide the combined READ-DATA. The PLO-CLK signals from the drives are converted to their single ended equivalents CK1 - CK4 by their line receivers. The CTL register bits US1 and US0 (for Unit Select) are used by the multiplexor U13 to select one of these signals to be the internal clock. The signal SEL-CLK is jumper selected from either the internal clock or its complement thus returning the PLO-CLK received from the drive back to the drive as a WRITE-CLOCK. This is required because the time delay of the round trip would introduce enough skew to make the interpretation of the WRITE-DATA at the drive unreliable if the data were clocked in with the source of PLO-CLK. The selection of internal CLOCK or CLOCK* is required because of an inconsistency in some drives. The signal cables also contain a single ended signal SEEK-COMPLETE*. The lines from the four drives are terminated and multiplexed again by U13 to provide SEEK-COMPLETE*.

IN THE MIDDLE - THE DISK DATA

Data is tranferred between the controller and disk in a bit serial format. The bit rates vary with the type of drive but are in the order of 5 MHz. The READ-DATA signal is delayed by one bit time in U11 to become READ-DLY and presented to the input of an octal shift register U58. Bytes of data are available in parallel at the outputs of the shift register every eight bit times. The broadside output of the shift register connects to the controllers internal shift register bus, the SR bus. During a disk read, the bytes are captured from the SR bus by the octal register U52 which then drives the data onto the computer's data output bus lines. U52 is clocked by the output of U11 which will have a rising edge in the center of the last bit time.

The READ-DLY signal is delayed one more bit time by U24 to provide READ-2-DLY*. This signal is ORed in U25-3 with a signal from U25-6 which will be false during a read. The READ-2-DLY is then shifted into a sixteen bit shift register formed by the two octal parts U41 and U42. This shift register has exclusive ORed feedback from its last bit to its first and two points in between, the specific connection of which results in the register accumulating the CRC check code for the read data. The control circuitry must compare the CRC computed as the data is read with the CRC code read from the disk immediately following the data. If the CRC computed during the read matches the CRC computed and stored during the write operation, the data is, in all probability, correct. The calculated CRC is available at the output of U42-12. This signal is connected to the left shift serial input of the shift register U58, so that the CRC may be brought into the last bit of the shift register by reversing the direction of shifting. This bit of the shift register is available on the Qh output and called WRITE-DATA. The output of U26-3 is the exclusive OR of WRITE-DATA (the computed CRC) and READ-2-DLY (the CRC read from the disk). A non-compare may be latched by U24, the SET-NE flip-flop.

Writing the disk uses much of the same hardware as read. Data read from memory on the data input bus is latched in U53 and held there until the end of a byte time of the hard disk at which point it is parallel loaded by the shift register from the SR bus. The serial output of the shift register, WRITE-DATA, is sent to the disk. During a write operation, U25-6 is enabled while all of the READ-DATA signals are zero so that the CRC register will be computing the CRC code for the write data. After all of the data has been written, the direction of the shift register is reversed causing the CRC to be shifted into the left shift input and out to the disk on the WRITE-DATA line. Besides writing data and CRC codes, the controller must also be capable of writing preambles (sequences of zeroes) and mark characters (in this case, an FFh). These constants are available on the SR bus from U57.

In addition to all of the above, the controller must identify the header field for the desired record before any data is read or written. The three bytes of header data to be matched are stored in the data registers CYLINDER, HEAD and SECTOR. The contents of these

registers may be gated onto the SR bus. During a read, the contents of a data register may be compared with data coming from the disk using the same hardware which is used for checking the CRC. The data register information is parallel loaded into the shift register at the start of a byte time. As the read data is shifting through READ-2-DLY, the shift register is shifted to the right, bringing the bits of the data word one at a time into the WRITE-DATA position. The two are compared and any non-compare is latched in SET-NE.

While all of the preceding discussion about reading and writing the disk discussed the capabilities of the shift register, the various sources of the SR bus, the CRC generator, the compare logic and the DMA data input and output registers, no mention of how and when any of these facilities are invoked was mentioned. The source for all of these control signals is a finite state machine (FSM) implemented with bipolar PROMs and latches. At the end of every byte time of the hard disk, a sixteen bit wide control word is read from the PROMs U47 & U48 into a register U30 and U31. The contents of these registers will control all of the other hardware for the next byte time as well as influencing which word will be read from the PROM next. The PROM address is made up of three components. The first is the operation code which is obtained from the OP2 - OP0 bits of the CTL register allowing up to eight instructions. The next field consists of K4 - K0, the current state, allowing a maximum of thirty two states. The last field is only one bit, the signal ALT. ALT allows the selection of one of two possible next states which shall be called normal and alternate. Thus every cycle of the machine is a conditional branch. There are sixteen bits of PROM output. One field of five bits contains the next state which become the current state K4 - K0 as it is loaded into the latch. Another field is latched to produce SA2 - SA0. This is the SR bus address and selects which of all of the SR bus sources discussed above will actually drive the bus during the next byte time. The SR bus address lines are decoded by (U39). The next field contain NEXTA - NEXTC. These are the address lines to the multiplexor U25 which selects which condition will generate ALT for the current cycle. There are four additional signals in the state vector. CRC* and LOAD control the operation of the shift register. The shift register normally shifts one bit to the right every bit time. If LOAD is true, the shift register will parallel load instead of shift in each bit 7 time. If CRC* is asserted, the shift register will shift to the left every bit time. READ and WRITE are buffered to become READ-GATE* and WRITE-GATE*. These are disk control signals which will be asserted just prior to the start of the next byte.

U33 is a four stage shift register which divides CLK by eight to form byte times. Its outputs, each of which is available in both true and complement form, make available a unique rising and falling edge for each bit time. One output, the signal Φa*, provides a rising edge used to clock the state register consisting of U30 and U31. This loads the control vector for the following byte time.

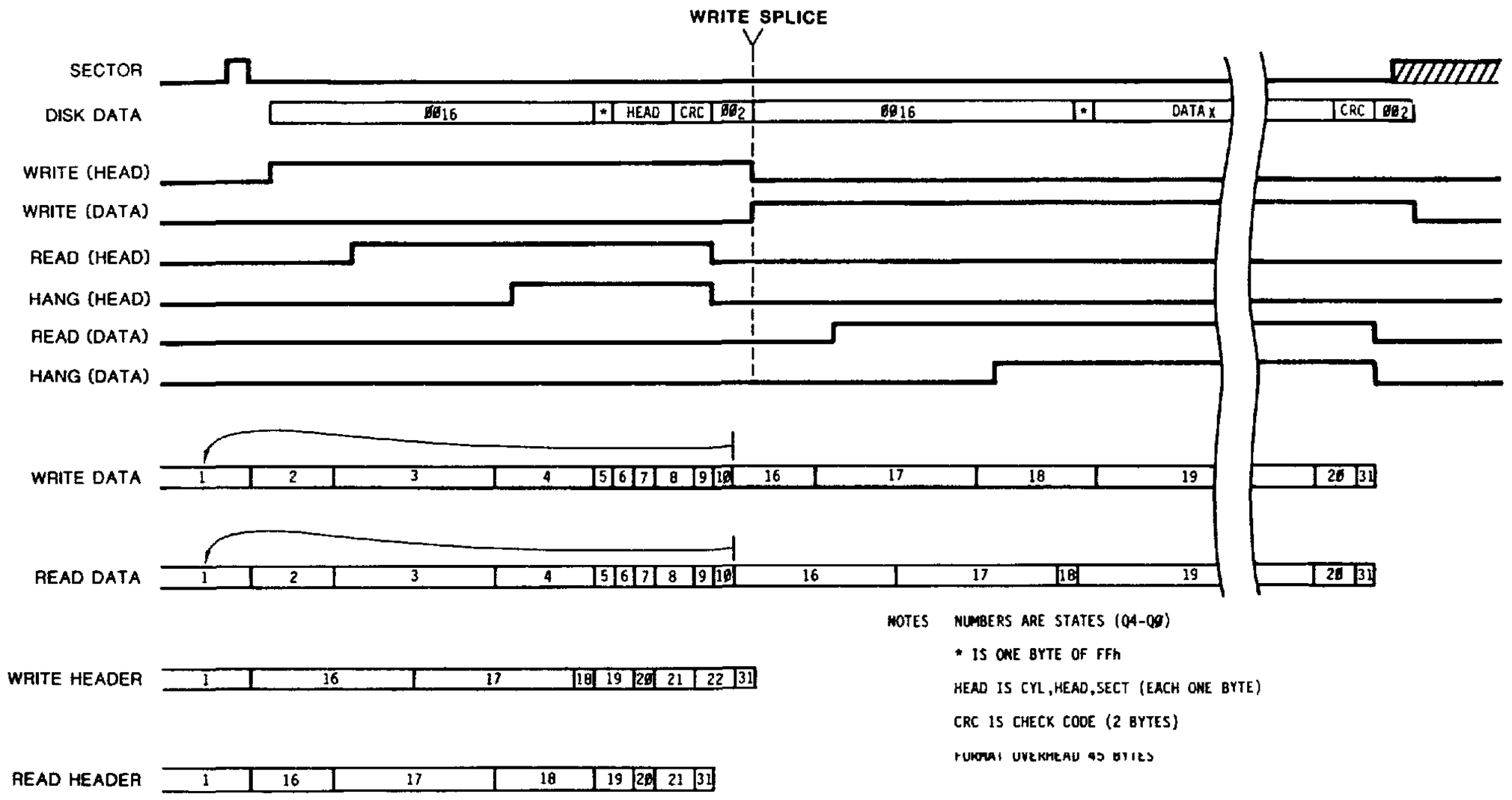
In the state register, bits K0 - K4 contain the present state number. NEXTA - NEXTC contain the branch condition which is used in selection of the following state. SA0 - SA2 contain the address of the SR bus source for the current byte time. All four of the remaining bits are reclocked into U32 by Φd which occurs one bit time prior to the start of the next byte. Two of the resulting outputs are READ and WRITE, the signals which control READ-GATE* and WRITE-GATE*. The other two lines are LOAD and CRC*. U34 qualifies LOAD with Φa and Φd. This will force the S1 control input of the shift register to be asserted during bit 7 if LOAD is true resulting in the shift register doing a parallel load at the start of the next byte time. The signal CRC* asserts S1 while negating S0 causing the shift register to reverse its direction for an entire byte time if CRC* is true.

The inputs of the STATE register all come from the outputs of a pair of PROMs U47 and U48. The address in the PROMs consists of three parts. The first field is OP2 - OP0, the current command. The next is K0 - K4, the current state number. The last input is a signal called ALT for alternate. Thus for every state of every command one of two exits is selected by the state of ALT. ALT originates at the output of U29, a multiplexor which selects one of eight conditions based on NEXTA - NEXTC. Its output, SET-ALT is reclocked by U32 to provide ALT. In designing the PROM, many states repeat for fixed numbers of byte times. This kind of action is attained by having a state transition to itself if ALT is false and advancing to the next state if ALT is true. Most of the ALT multiplexor inputs are of this form. These include N1, N2, N4, N8 and NX. N1 is always true. N2 - N8 allow states with the implied durations. NX is switch selected to one of N128, N256, N512, N1024 or N2048 depending on the disk formatting. The remaining multiplexor inputs are RUN, which allows 0 states to wait for the machine to be turned on, NE which allows it to verify a header or CRC match, and SEC which allows it to wait for the the next SECTOR* from the disk. The byte times all come from the byte counter U27 and U28 which counts once each byte time due to Φb* and is reset at the start of each new state by ALT and Φa* in U46. SEC is the SECTOR* signal from the disk latched by U18 to hold it presence for at least one byte time.

The control software should verify a READY status from the selected drive before starting an automatic operation for two reasons. First, the finite state machine is clocked by the disk and so will hang if a non-existent or not ready disk is selected. The second is that the TIME-OUT mechanism which is used to halt the machine if, for example, a header is not found uses INDEX* from the selected drive. Index clocks one of the counters in U27. This counter is held in a cleared state until RUN and SEEK-COMPLETE are true by U46. If a operation fails to complete in two passes of the index from the time it starts, TIME-OUT will be generated clearing and halting the machine.

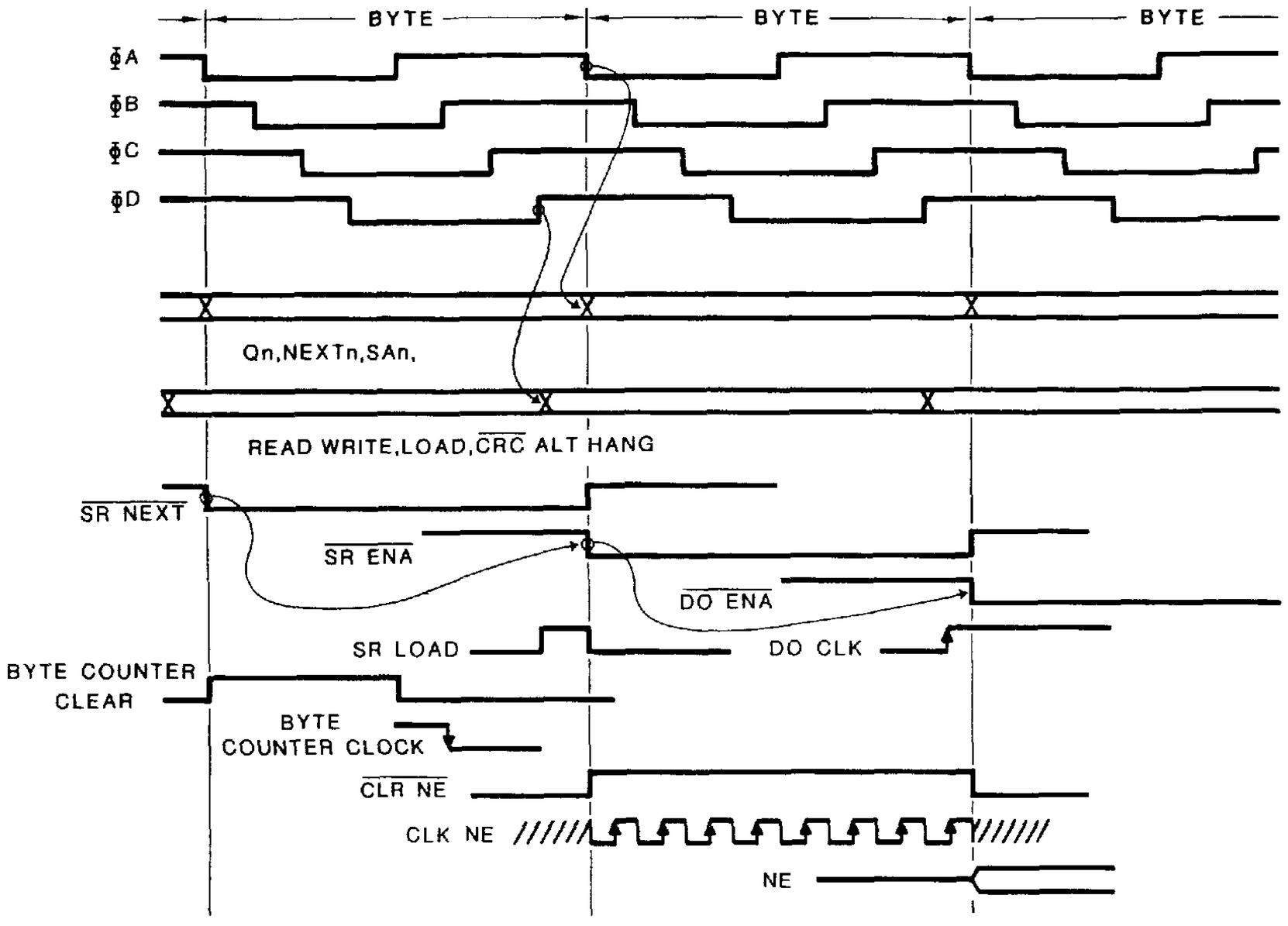
The PROM code implements the track formats/timing diagrams as shown below in Figure 1. This diagram relates the various states of the four major operations to the position of the disk (i.e. relative to sector pulses and recorded data). Note that all of the functions wait for the start of a sector. READ HEADER or WRITE HEADER will proceed to completion on finding a header. READ DATA and WRITE DATA will loop and wait for the next sector if the current sector header does not match the desired cylinder, head and sector number and have a current CRC check code.

Figure 1. DISK 2: TRACK FORMAT AND STATE TIMING

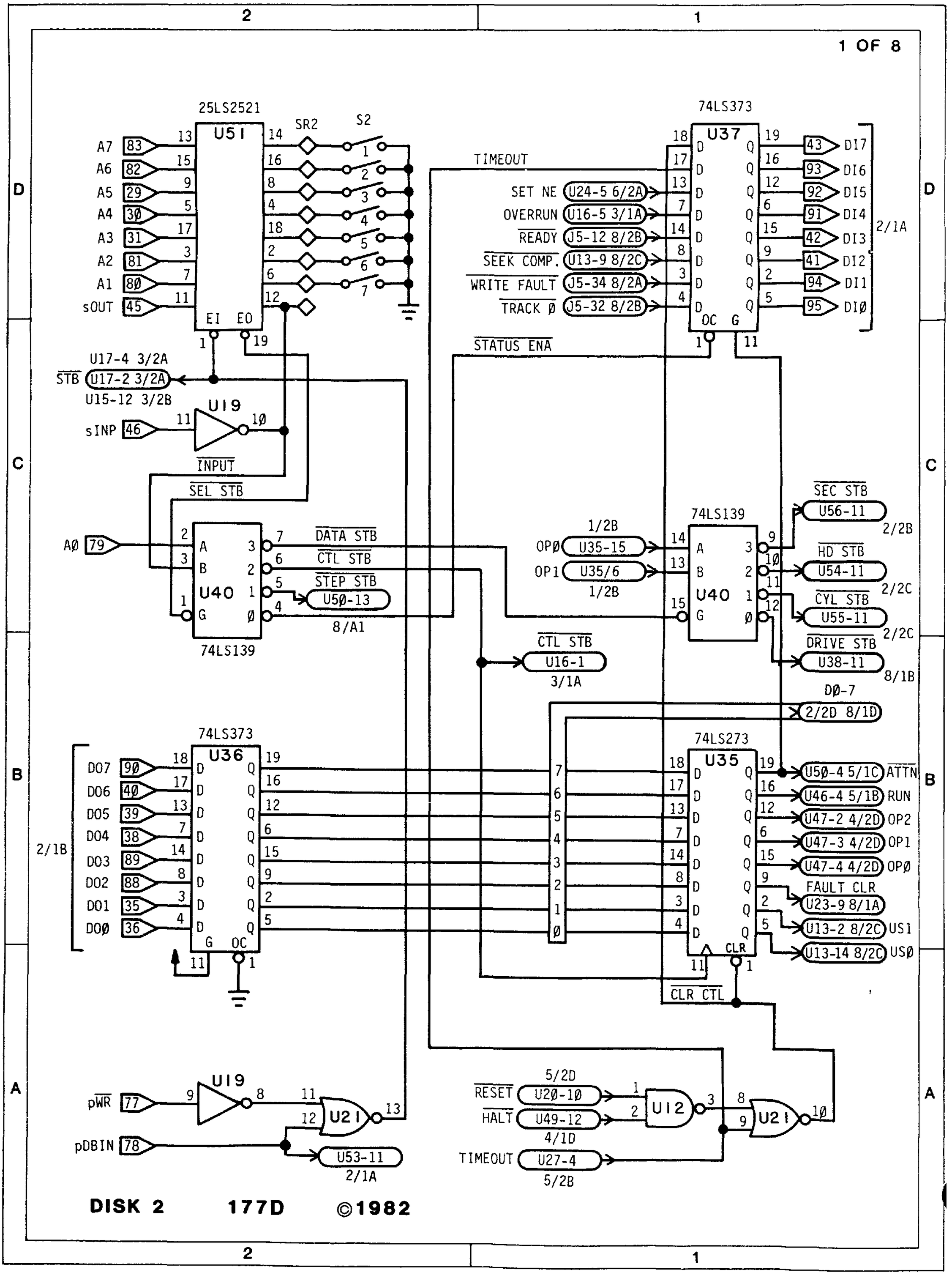


Not all signals change state on byte boundaries. Figure 2 shows the timing of various signals relative to the clock phases. Note that the state responsible for a given action is generally about one byte ahead of the data on the disk. This is because writing a byte of data to the disk or checking a byte of data read from the disk requires that data be read from a register during the preceding byte time and loaded into the shift register at the very start of the following byte time. In the singular case of reading data from disk to memory, the command in the current byte causes the byte of data read during the following byte to be enabled onto the SR bus from which it will be captured just prior to the start of the third byte by the DO register. During the third byte time, the data will be sent to memory.

Figure 2. DISK 2: TIMING OF VARIOUS SIGNAL RELATIVE TO THE CLOCK PHASES.



HARDWARE SECTION

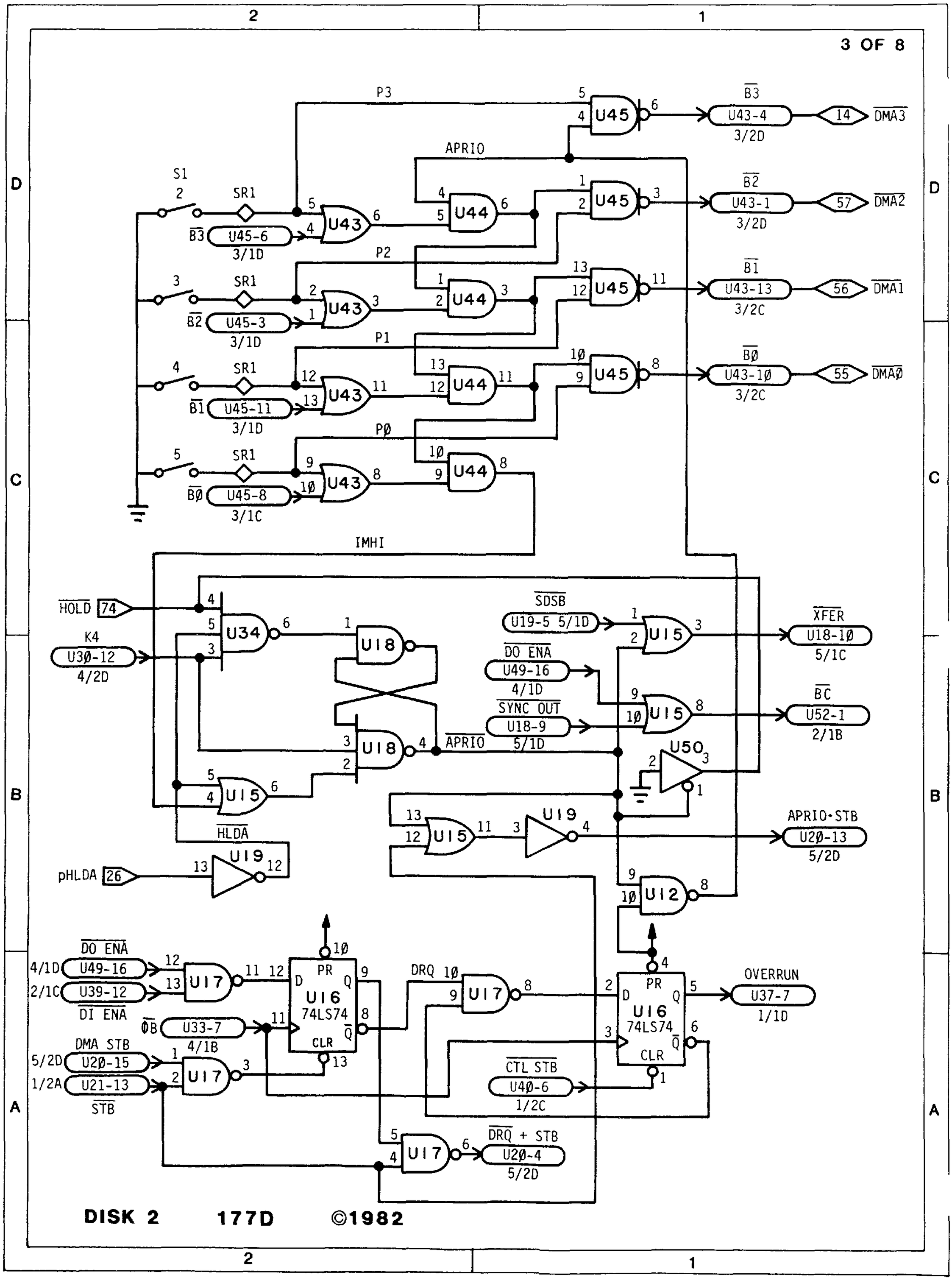

25LS2521
U51
A7 83
A6 82
A5 29
A4 3Ø
A3 31
A2 81
A1 8Ø
sOUT 45
EI EO
SR2
S2
74LS373
U37
TIMEOUT
SET NE U24-5 6/2A
OVERRUN U16-5 3/1A
READY J5-12 8/2B
SEEK COMP. U13-9 8/2C
WRITE FAULT J5-34 8/2A
TRACK Ø J5-32 8/2B
DI7 DI6 DI5 DI4 DI3 DI2 DI1 DIØ
2/1A
STATUS ENA
U17-4 3/2A
STB U17-2 3/2A
U15-12 3/2B
U19
sINP 46
INPUT
SEL STB
AØ 79
U40
74LS139
DATA STB
CTL STB
STEP STB
U5Ø-13
8/A1
1/2B
OPØ U35-15
OP1 U35/6
1/2B
SEC STB
U56-11
2/2B
HD STB
U54-11
2/2C
CYL STB
U55-11
2/2C
DRIVE STB
U38-11
8/1B
CTL STB
U16-1
3/1A
DØ-7
2/2D 8/1D
74LS373
U36
DO7 9Ø
DO6 4Ø
DO5 39
DO4 38
DO3 89
DO2 88
DO1 35
DOØ 36
2/1B
74LS273
U35
ATTN
RUN
OP2
OP1
OPØ
FAULT CLR
U23-9 8/1A
US1
USØ
CLR CTL
pWR 77
pDBIN 78
U21
U53-11
2/1A
5/2D
RESET U2Ø-1Ø
HALT U49-12
4/1D
U12
TIMEOUT U27-4
5/2B
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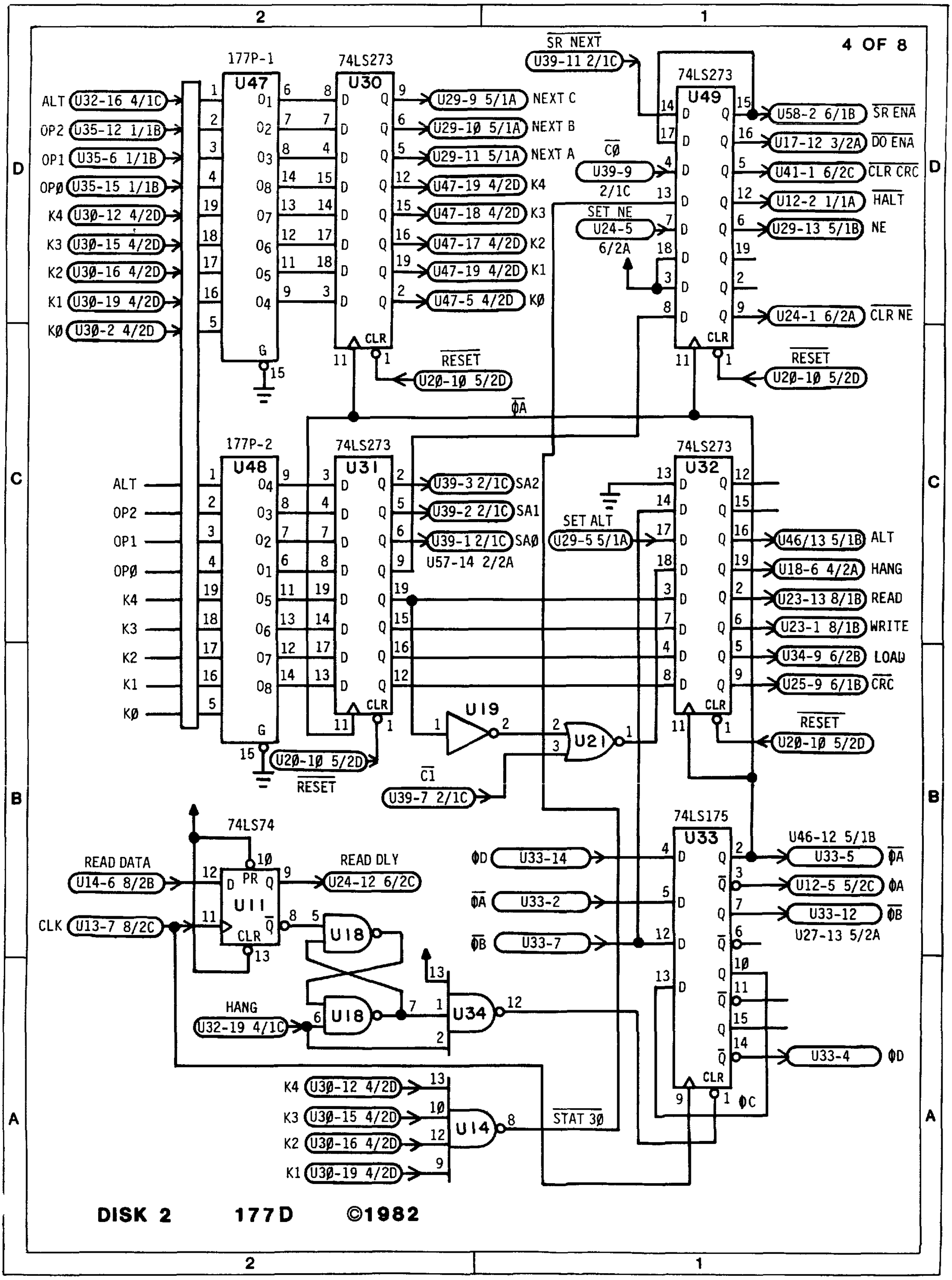
◇ = SIP PULL-UP RESISTOR = V_{CC} SR X

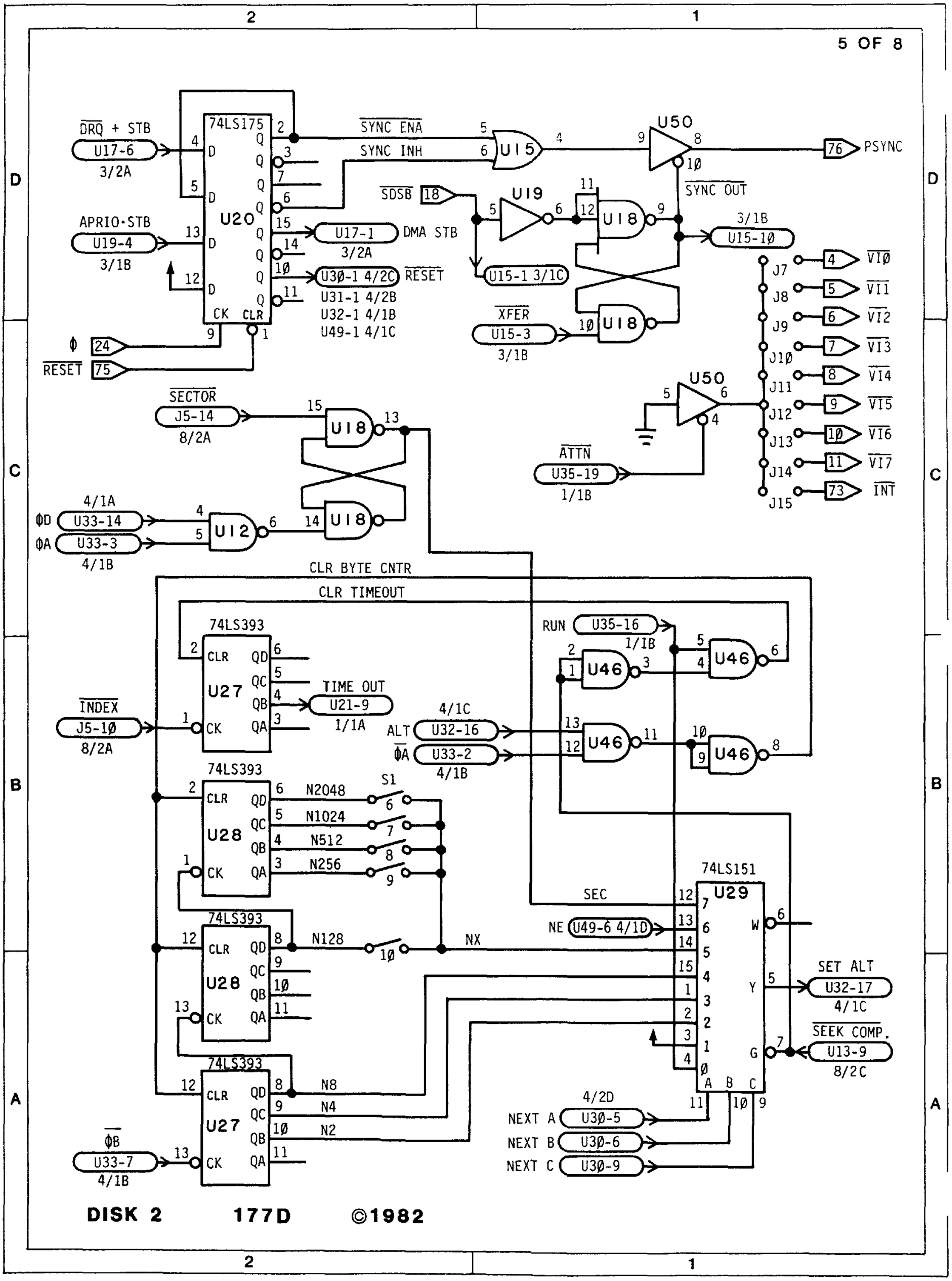
SØ-7 6/1B
DØ-7 1/1B
CYL STB U4Ø-11 1/1C
HD STB U4Ø-1Ø 1/1C
SEC STB U4Ø-9 1/1C
SAØ U31-6 4/2C
SA1 U31-5 4/2C
SA2 U31-2 4/2C
K ENA
CYL ENA
HD ENA
SEC STB
SØ-S7
ØD U33-14 8/2D
CLK U25-11 8/2D
DI7 43
DI6 93
DI5 92
DI4 91
DI3 42
DI2 41
DI1 94
DIØ 95
1/2D
pDBIN 78
4/1B U21-3 CI
U49-4 CØ 4/1D
U49-14 SR NEXT 4/1D
DI ENA U17-13 3/2A
9Ø DO7
4Ø DO6
39 DO5
38 DO4
89 DO3
88 DO2
35 DO1
36 DOØ
1/2B
U15-8 BC 3/1B

U54 74LS374
U55 74LS374
U56 74LS374
U57 74LS373
U39 74LS138
U12
U11 74LS74
U52 74LS374
U53 74LS373

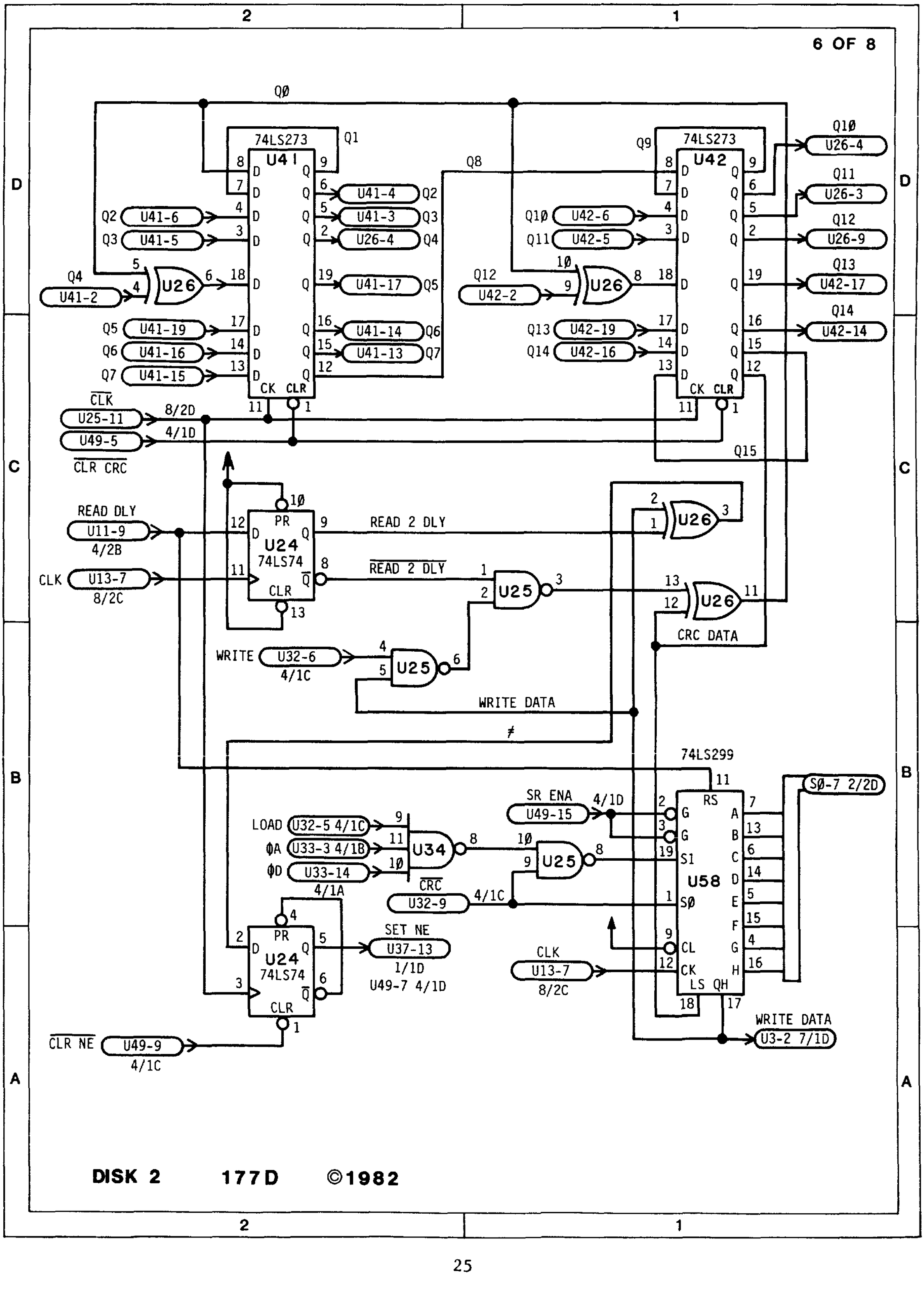
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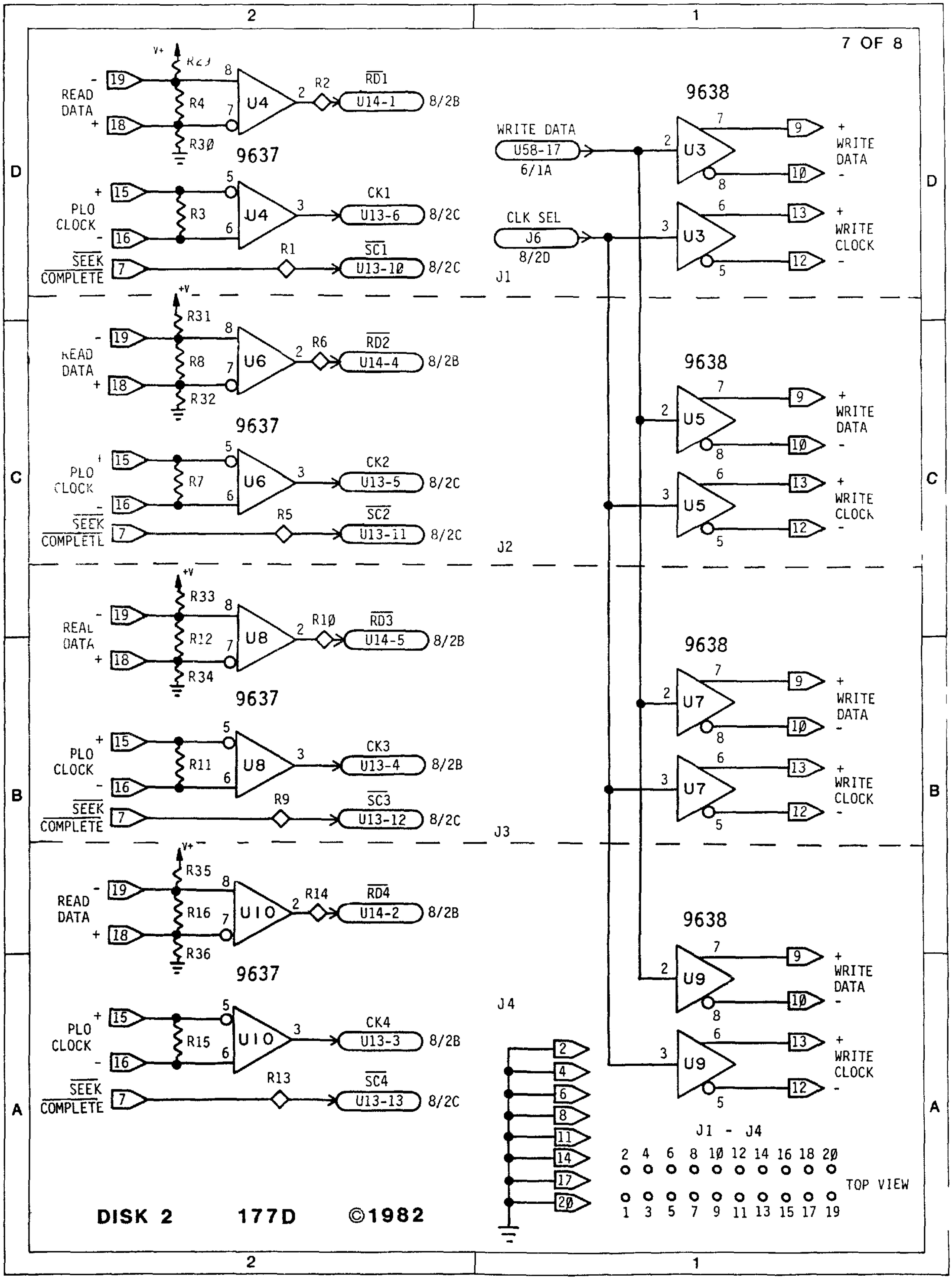

P3
APRIO
S1
SR1
B3
U45-6
3/1D
U43
U44
U45
B3
U43-4
3/2D
DMA3
B2
U43-1
3/2D
DMA2
P2
B2
U45-3
3/1D
B1
U43-13
3/2C
DMA1
P1
B1
U45-11
3/1D
BØ
U43-1Ø
3/2C
DMAØ
PØ
BØ
U45-8
3/1C
IMHI
HOLD 74
K4
U3Ø-12
4/2D
U34
U18
U15
HLDA
pHLDA 26
U19
SDSB
U19-5 5/1D
DO ENA
U49-16
4/1D
SYNC OUT
U18-9
5/1D
APRIO
XFER
U18-1Ø
5/1C
BC
U52-1
2/1B
U50
APRIO·STB
U2Ø-13
5/2D
U12
DO ENA
4/1D U49-16
2/1C U39-12
DI ENA
ØB U33-7
4/1B
DMA STB
5/2D U2Ø-15
1/2A U21-13
STB
U17
PR
D
Q
U16
74LS74
CLR
DRQ 1Ø
OVERRUN
U37-7
1/1D
CTL STB
U4Ø-6
1/2C
DRQ + STB
U2Ø-4
5/2D
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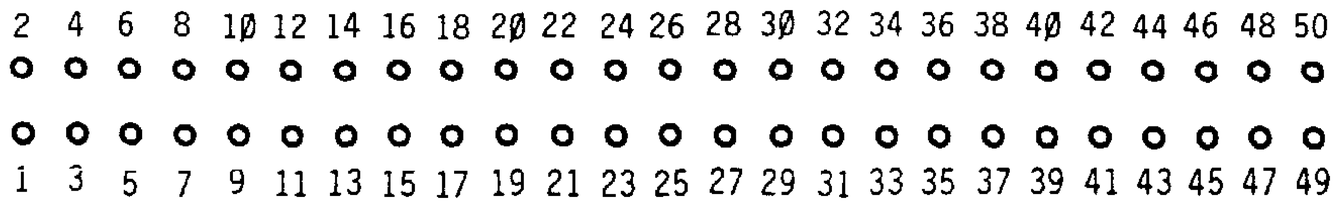

4 OF 8
177P-1
U47
74LS273
U30
74LS273
U49
SR NEXT
U39-11 2/1C
ALT U32-16 4/1C
OP2 U35-12 1/1B
OP1 U35-6 1/1B
OPØ U35-15 1/1B
K4 U3Ø-12 4/2D
K3 U3Ø-15 4/2D
K2 U3Ø-16 4/2D
K1 U3Ø-19 4/2D
KØ U3Ø-2 4/2D
U29-9 5/1A NEXT C
U29-1Ø 5/1A NEXT B
U29-11 5/1A NEXT A
U47-19 4/2D K4
U47-18 4/2D K3
U47-17 4/2D K2
U47-19 4/2D K1
U47-5 4/2D KØ
CØ
U39-9 2/1C
SET NE
U24-5 6/2A
U58-2 6/1B SR ENA
U17-12 3/2A DO ENA
U41-1 6/2C CLR CRC
U12-2 1/1A HALT
U29-13 5/1B NE
U24-1 6/2A CLR NE
RESET
U2Ø-1Ø 5/2D
ØA
177P-2
U48
74LS273
U31
74LS273
U32
U39-3 2/1C SA2
U39-2 2/1C SA1
U39-1 2/1C SAØ
U57-14 2/2A
SET ALT
U29-5 5/1A
U46/13 5/1B ALT
U18-6 4/2A HANG
U23-13 8/1B READ
U23-1 8/1B WRITE
U34-9 6/2B LOAD
U25-9 6/1B CRC
U19
U21
CI
U39-7 2/1C
74LS74
U11
READ DATA
U14-6 8/2B
READ DLY
U24-12 6/2C
CLK U13-7 8/2C
U18
HANG
U32-19 4/1C
U34
74LS175
U33
ØD U33-14
ØA U33-2
ØB U33-7
U46-12 5/1B
U33-5 ØA
U12-5 5/2C ØA
U33-12 ØB
U27-13 5/2A
U33-4 ØD
ØC
K4 U3Ø-12 4/2D
K3 U3Ø-15 4/2D
K2 U3Ø-16 4/2D
K1 U3Ø-19 4/2D
U14
STAT 3Ø
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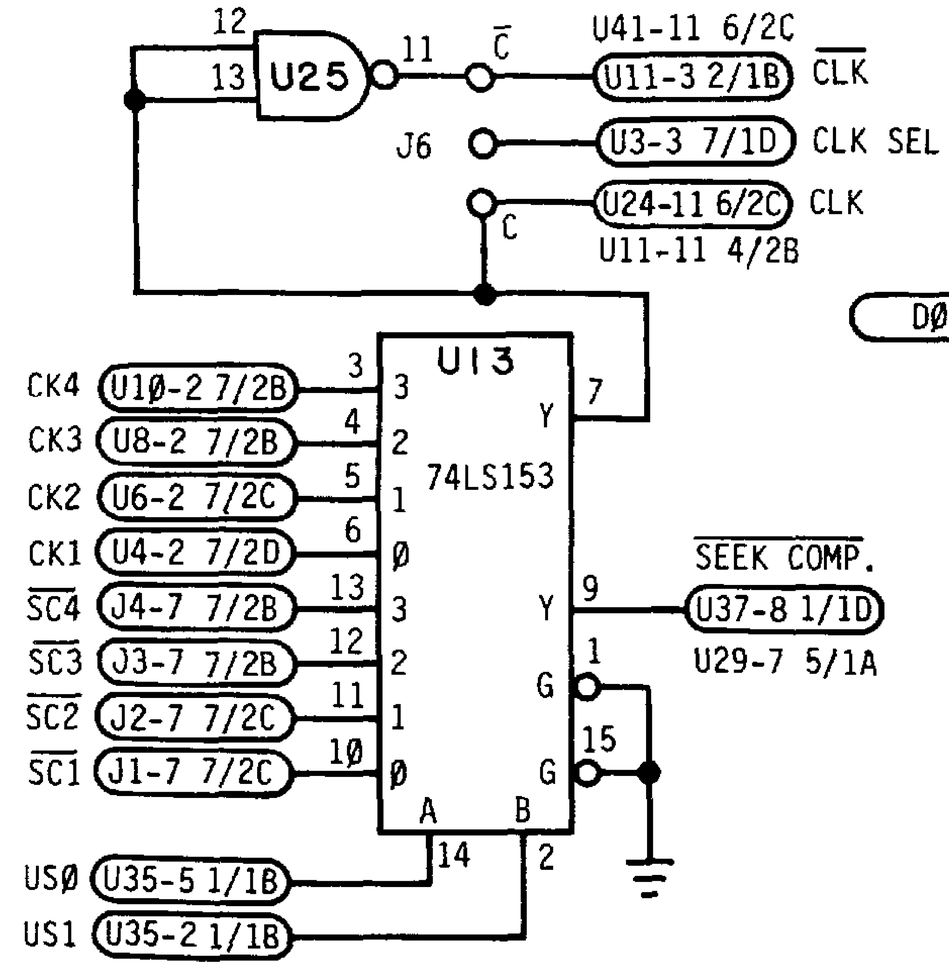
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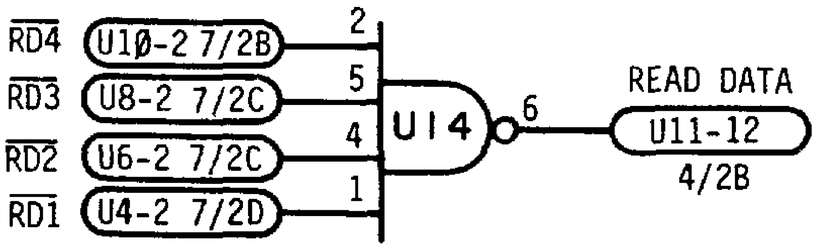

Q0
74LS273
U41
Q1
Q2
Q3
Q4
Q5
Q6
Q7
U41-4
U41-3
U26-4
U41-17
U41-14
U41-13
U41-2
U41-6
U41-5
U41-19
U41-16
U41-15
U26
CK CLR
CLK
U25-11
U49-5
CLR CRC
8/2D
4/1D
Q8
74LS273
U42
Q9
Q10
Q11
Q12
Q13
Q14
Q15
U26-4
U26-3
U26-9
U42-17
U42-14
U42-2
U42-6
U42-5
U42-19
U42-16
U26
CK CLR
READ DLY
U11-9
4/2B
CLK
U13-7
8/2C
U24
74LS74
PR
CLR
READ 2 DLY
U25
U26
CRC DATA
WRITE
U32-6
4/1C
WRITE DATA
≠
74LS299
U58
RS
LS QH
SR ENA
U49-15
4/1D
SØ-7 2/2D
LOAD
U32-5 4/1C
ØA
U33-3 4/1B
ØD
U33-14
4/1A
U34
CRC
U32-9
4/1C
SET NE
U37-13
1/1D
U49-7 4/1D
CLR NE
U49-9
4/1C
WRITE DATA
U3-2 7/1D
DISK 2
177D
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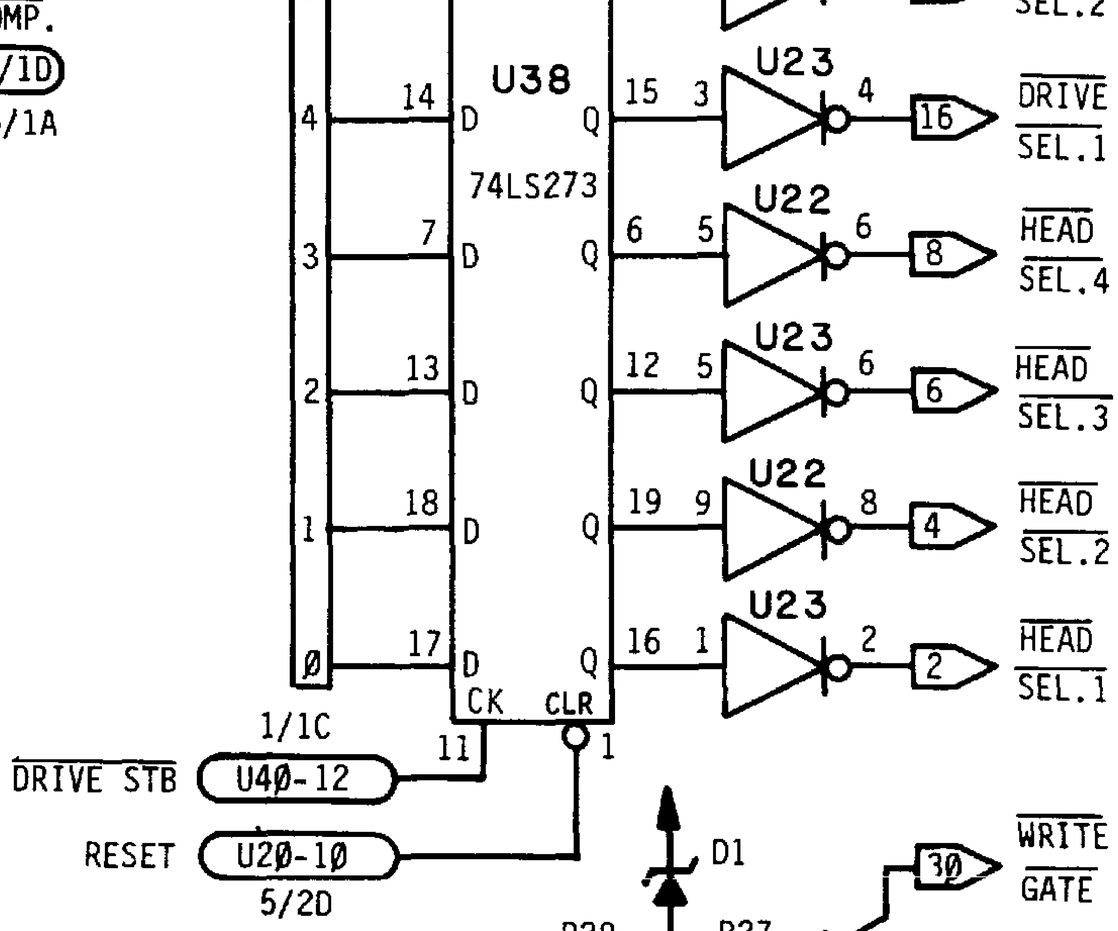

7 OF 8
2
1
D
C
B
A
9637
READ DATA
PLO CLOCK
SEEK COMPLETE
R4
R3
R30
R8
R7
R32
R12
R11
R34
R16
R15
R36
U4
U6
U8
U10
R2
R6
R1
R5
R10
R14
R13
RD1
RD2
RD3
RD4
SC1
SC2
SC3
SC4
U14-1
U14-4
U14-5
U14-2
U13-6
U13-5
U13-12
U13-3
U13-13
U13-10
U13-11
U13-4
8/2B
8/2C
J1
J2
J3
J4
9638
WRITE DATA
U58-17
6/1A
CLK SEL
J6
8/2D
U3
U5
U7
U9
+ WRITE DATA -
+ WRITE CLOCK -
J1 - J4
2 4 6 8 10 12 14 16 18 20
1 3 5 7 9 11 13 15 17 19
TOP VIEW
DISK 2 177D ©1982

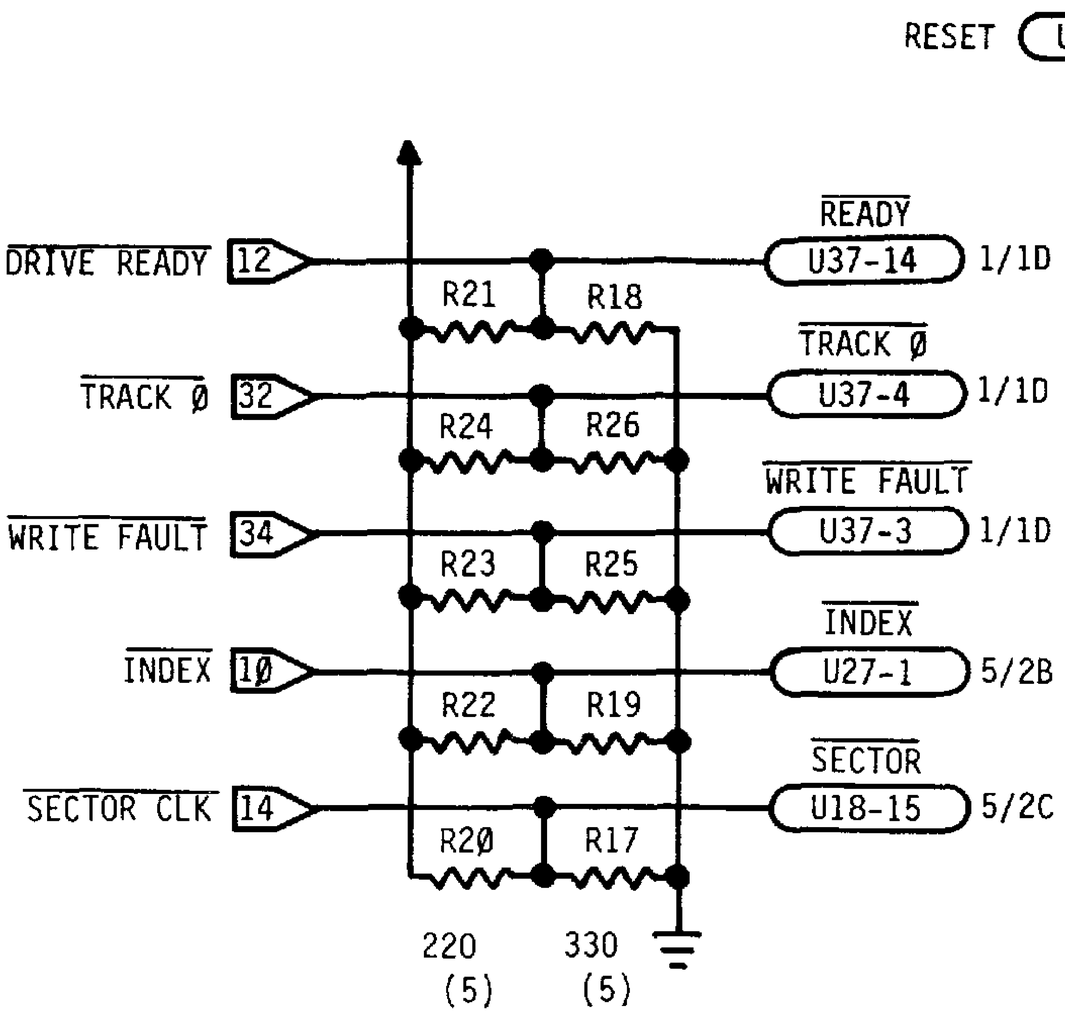


GROUND - PINS 1-37 ODD, 41,44,47,50









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PARTS LIST

SEMICONDUCTORS

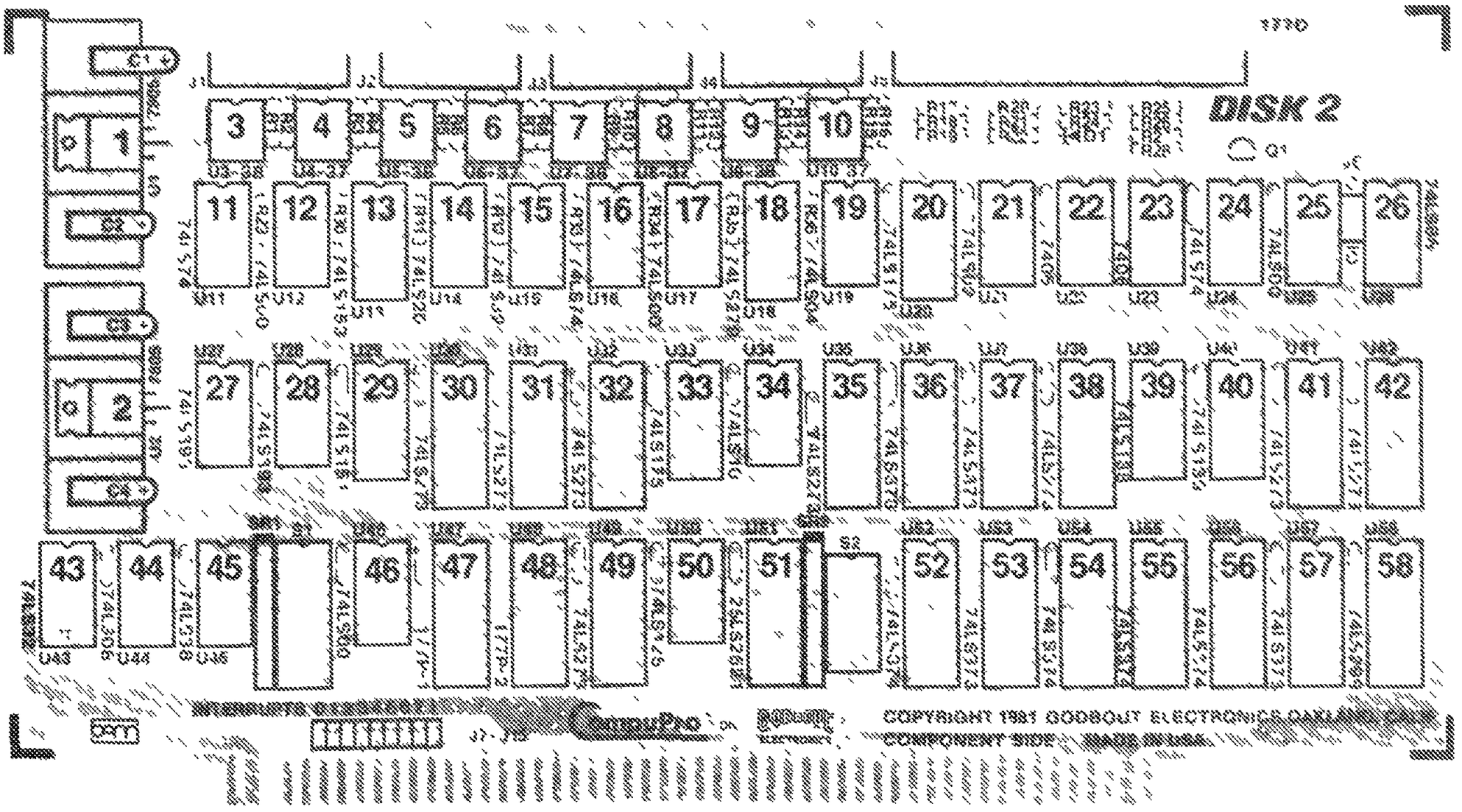
UNIT#	PART#
U1	7805
U2	7805
U3	9638
U4	9637
U5	9638
U6	9637
U7	9638
U8	9637
U9	9638
U10	9637
U11	74LS74
U12	74LS00
U13	74LS153
U14	74LS20
U15	74LS32
U16	74LS74
U17	74LS00
U18	74LS279
U19	74LS04
U20	74LS175
U21	74LS02
U22	7406/16
U23	7406/16
U24	74LS74
U25	74LS00
U26	74LS86
U27	74LS393
U28	74LS393
U29	74LS151
U30	74LS273
U31	74LS273
U32	74LS273

UNIT#	PARTS#
U33	74LS175
U34	74LS10
U35	74LS273
U36	74LS373
U37	74LS373
U38	74LS273
U39	74LS138
U40	74LS139
U41	74LS273
U42	74LS273
U43	74LS32
U44	74LS08
U45	74LS38
U46	74LS00
U47	177P-1
U48	177P-2
U49	74LS273
U50	74LS125
U51	25LS2521
U52	74LS374
U53	74LS373
U54	74LS374
U55	74LS374
U56	74LS374
U57	74LS373
U58	74LS299
Q1	2N2222

DIODE	PART#
D1	1N747

RESISTORS

	VALUE
R1,2	4.7K Ohm
R3,4	110 Ohm
R5,6	4.7K Ohm
R7,8	110 Ohm
R9,10	4.7K Ohm
R11,12	110 Ohm
R13,14	4.7K Ohm
R15,16	110 Ohm
R17-19	330 Ohm
R20-24	220 Ohm
R25-27	330 Ohm
R28	68 Ohm
R29-36	1K Ohm
SR1	2.7K-5 SIP
SR2	2.7K-5 SIP



COMPONENT LAYOUT

ABOUT SELECTOR CHANNEL

The SELECTOR CHANNEL is an IEEE 696/S-100 compatible board which is designed to provide all of the bus address, status and control strobes required by a temporary master (DMA device) designed to use it. The combination of a selector channel and companion device controller are the functionally equivalent to a single board temporary master. The hardware on the SELECTOR CHANNEL however may be shared by a number of device controllers.

TECHNICAL OVERVIEW

The SELECTOR CHANNEL has a single I/O port through which the CPU or other bus master may program the identification of the device controller to be supported as well as the addresses of and types of cycles to be generated. When the selected device is granted access to the bus, the SELECTOR CHANNEL will exchange mastership of all of the address, status and control lines with the CPU and generate cycles as requested by the device controller. In addition, the SELECTOR CHANNEL has a BOOT function which may be used to initialize the system from one of the devices whose controllers share the SELECTOR CHANNEL.

SWITCH SETTINGS AND OPTION SELECTION

The SELECTOR CHANNEL's I/O port may be addressed anywhere in the 00h - FFh range using switches SW1-3 through SW1-10. SW1-1 is used to enable the on board BOOT EPROM. SW1-2 is used to enable one wait state minimum for all DMA cycles generated by the SELECTOR CHANNEL. Wait states may also be requested in software when the SELECTOR CHANNEL is programmed and the SELECTOR CHANNEL will honor wait states requested by bus slaves during temporary master cycles which is supports.

In addition to Switch 1, the SELECTOR CHANNEL has a number of options which are controlled by the installation of push-on jumpers. J1 - J4 enable one through four wait states respectively for the on board EPROM. As the SELECTOR CHANNEL is shipped, J4 is connected by a trace to allow maximum access for slow EPROMS. If fewer EPROM wait states are required, this trace may be cut and a jumper may be installed in one of the other locations. These waits are only generated when initially booting a system with the EPROM, so the insertions of waits will not adversely affect system operation.

The SELECTOR CHANNEL uses a 2716 type EPROM which has a capacity of 2 K bytes. Since this capacity is greater than required for most boot operations, jumpering is provided which will allow the space to be divided into several smaller code segments and the selection of one of them. This will enable a single standard EPROM to hold boot code for a number of device controllers and configurations. Jumper locations J8 through J10 allow the corresponding address lines of the EPROM to be tied to the corresponding bus address lines providing a larger address range or to be tied high and low in any combination selecting a section of code within the 2716.

PROGRAMMING THE SELECTOR CHANNEL

The first step in programming the SELECTOR CHANNEL is to do a read from its I/O port. The read operation will disable the SELECTOR CHANNEL during the following four write cycles preventing it from supporting any temporary master cycles while it has a partial program. No data is returned by the read. The SELECTOR CHANNEL powers up in this disabled state.

The SELECTOR CHANNEL must then be sent a sequence of exactly four bytes. They are as follows:

(1) A23-A16 The extended address byte
(2) A15-A8 The high address byte
(3) A7-A0 The low address byte
(4) MODE The mode control byte

Three address bytes are required because the SELECTOR CHANNEL addresses the full 24 bit extended address range of the IEEE 696/S-100 bus. An extended address byte must be sent even if the memory or I/O devices which will be addressed are going ignore its contents. Example: to send address 0FFFF0 the bytes in order would be 0F, FF, and F0. The mode byte is made up of two parts. The bits in the upper nibble control the mode as follows:

BIT	NAME	FUNCTION
D7	W/R*	Write / Read relative to the bus slave (memory or I/O port).
D6	IO/M*	IO / Memory - controls the status of cycles generated. Also, memory addresses are incremented on multi-byte transfers. I/O addresses are not.
D5	U/D*	Up / Down - A memory address may be counted up or down after each byte.
D4	WAIT	WAIT - Imposes a single wait state for each bus cycles supported.

The lower nibble of the MODE byte contains the identification of the controller to be supported. This nibble should contain the complement of the device priority.

BIT	NAME	FUNCTION
D3	DMA 3*	Select complement of DMA device that
D2	DMA 2*	Selector Channel will work with.
D1	DMA 1*	
D0	DMA 0*	

Once the SELECTOR CHANNEL has been programmed, it will respond to all cycles requested by the selected device. The SELECTOR CHANNEL function is independent of whether the device does individual cycles or burst mode transfers. The SELECTOR CHANNEL may also

support a sequence of burst mode transfers to successive locations without reprogramming. If a controller function for which the SELECTOR CHANNEL may have transferred one or more bytes of data is going to be repeated, the SELECTOR CHANNEL must be reprogrammed. The SELECTOR CHANNEL may be disabled at any time by reading its I/O port.

THEORY OF OPERATION

The octal comparator U26 decodes the SELECTOR CHANNEL's I/O port address qualified by I/O status to provide BD-SEL*. The bus reset signal RESET* or BD-SEL* in conjunction with the read strobe pDBIN will disable the SELECTOR CHANNEL by clearing the counter U10. U10 will count write cycles since it is clocked by BD-SEL* and pWR*, the bus write strobe. After four write cycles, CLEAR* which is obtained from the Qd output of U10 will hang U10 by disqualifying its clock and enable the SELECTOR CHANNEL to respond to the selected controller.

All data sent to the SELECTOR CHANNEL is loaded into the MODE register U17. The output of U17 is connected in parallel to the least significant byte of address counter formed by the four bit counters U6 and U16. Similarly, the output of the low byte of the counter provides the input to the high byte counter U3 and U13 which in turn provides the input to the extended byte counter U4 and U14. Thus the MODE register and three bytes of address counter form a four stage, eight bit wide shift register. The counters are set to load rather than count whenever the SELECTOR CHANNEL is not actually involved in a temporary master cycle. A LD-STB derived from BD-SEL* and pWR* clocks the address counters as well the MODE register causing the shifts. The low, high and extended address counters have associated bus drivers U25, U22 and U23 respectively, which are enabled during master cycles.

The IO/M* and W/R* status bits of the MODE register are decoded by U7 along with the generalized strobes. The resulting status is driven onto the bus by U18 during master cycles while the strobes are output by U24.

The XFER1 flip-flop (U5) is used to detect that the selected controller has obtained the bus. The flip-flop will set at the first falling edge of the bus clock following the combination of HOLD*, HLDA and IMHI all being true. IMHI indicates a match between the priority of the device holding the bus and the target device whose priority number is stored in the low nibble of the MODE register. When XFER1 sets, it indicates the start of the overlap period during which the permanent and temporary masters both drive the command lines. If XFER1 is set, XFER2 in U12 will set at the next rising edge of the clock. XFER2 enables the balance of the bus drivers and disables the remainder of the masters bus drivers. Once XFER2 has set, the SELECTOR CHANNEL will wait for cycles to be requested by the controller.

When the device controller is ready to transfer a byte of data, it will generate the bus signal pSYNC. When the SELECTOR CHANNEL detects pSYNC, it will start a bus cycle by generating a pSTVAL* signal when pSYNC is high and 0 is low. The read or write strobe will be asserted with the next rising edge of 0 and stay asserted for one clock cycle unless a DMA wait or external wait is requested by having a WRQ signal asserted. The strobe will go away one clock cycle after the WRQ signal goes low. The BC signal will be asserted on the next clock after the read or write strobe is asserted and then go away after the strobe is released. The falling edge of BC will step the address counters to the next DMA address for the next data transfer.

When the controller is finished with the bus, it will remove the signal HOLD*. The following rising edge of the clock will remove XFER2 and the negative edge after that will remove XFER1.

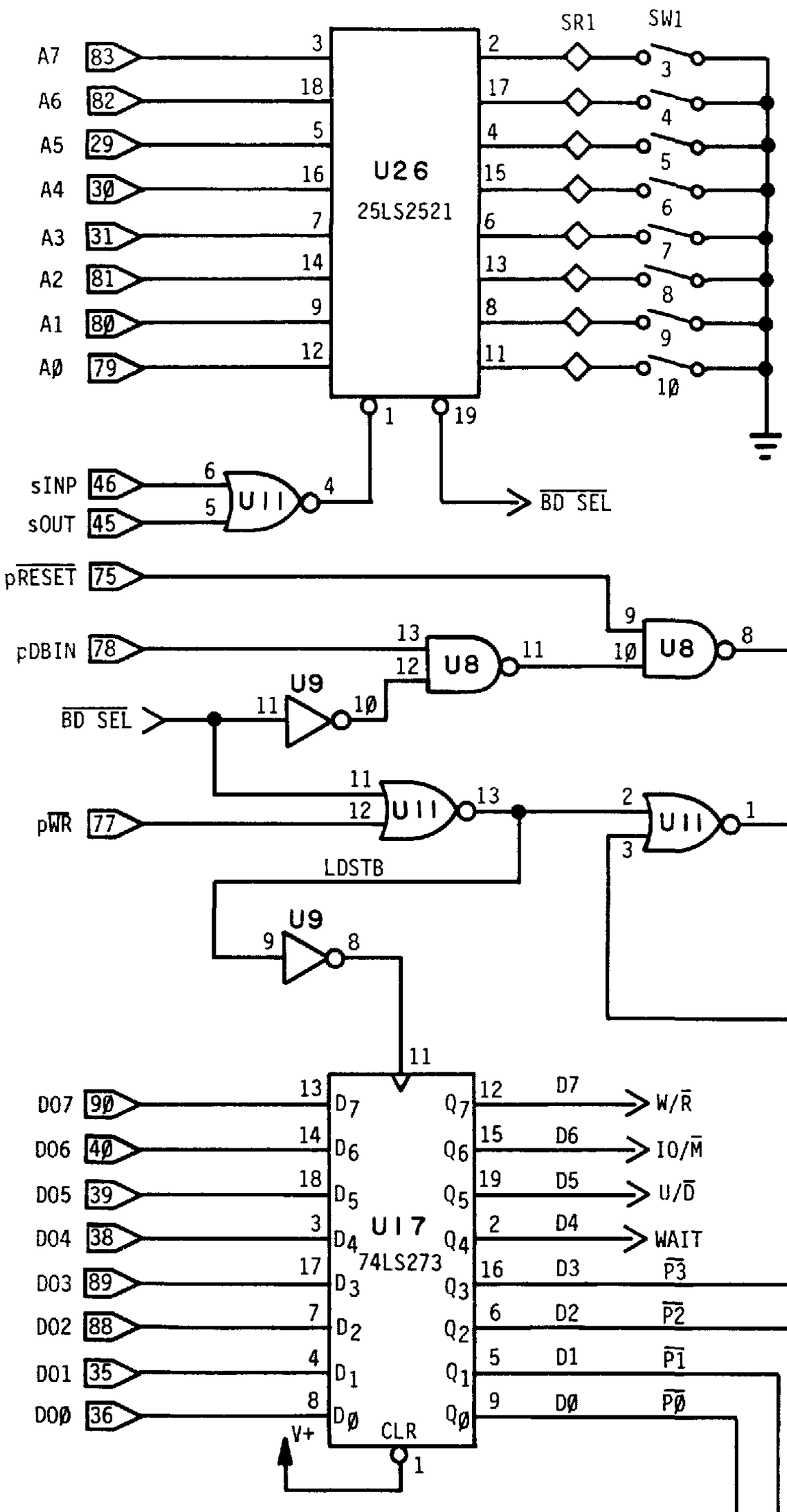
The on board boot EPROM may share address space with RAM memory which responds to PHANTOM*. The BOOT latch (U30) is enabled at RESET* and will assert PHANTOM* (if SW1-1 is ON) on all CPU memory reads, disabling the RAM and enabling the on board EPROM. PHANTOM* is not asserted during temporary master operations, thus allowing data transfers between the device controller and RAM. The BOOT latch will be reset the first time the CPU writes to RAM. The EPROM (U27) receives its addresses directly from the bus. The data driver (U28) places it output on the data input bus during memory reads when the boot is enabled. The counter U19 implements wait states by pulling RDY low for a selected number of clock cycles of strobe.

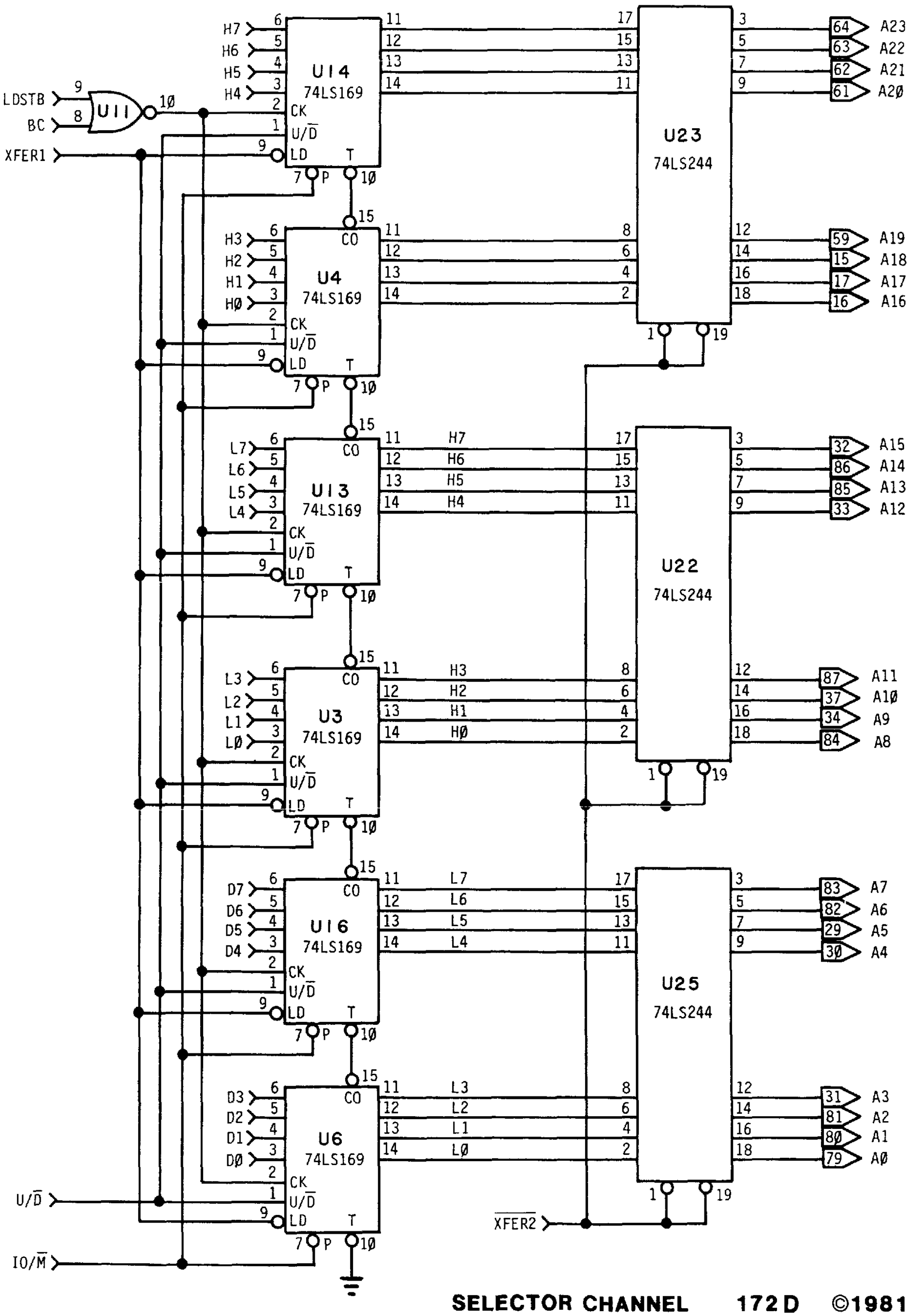
HARDWARE SECTION

SELECTOR CHANNEL

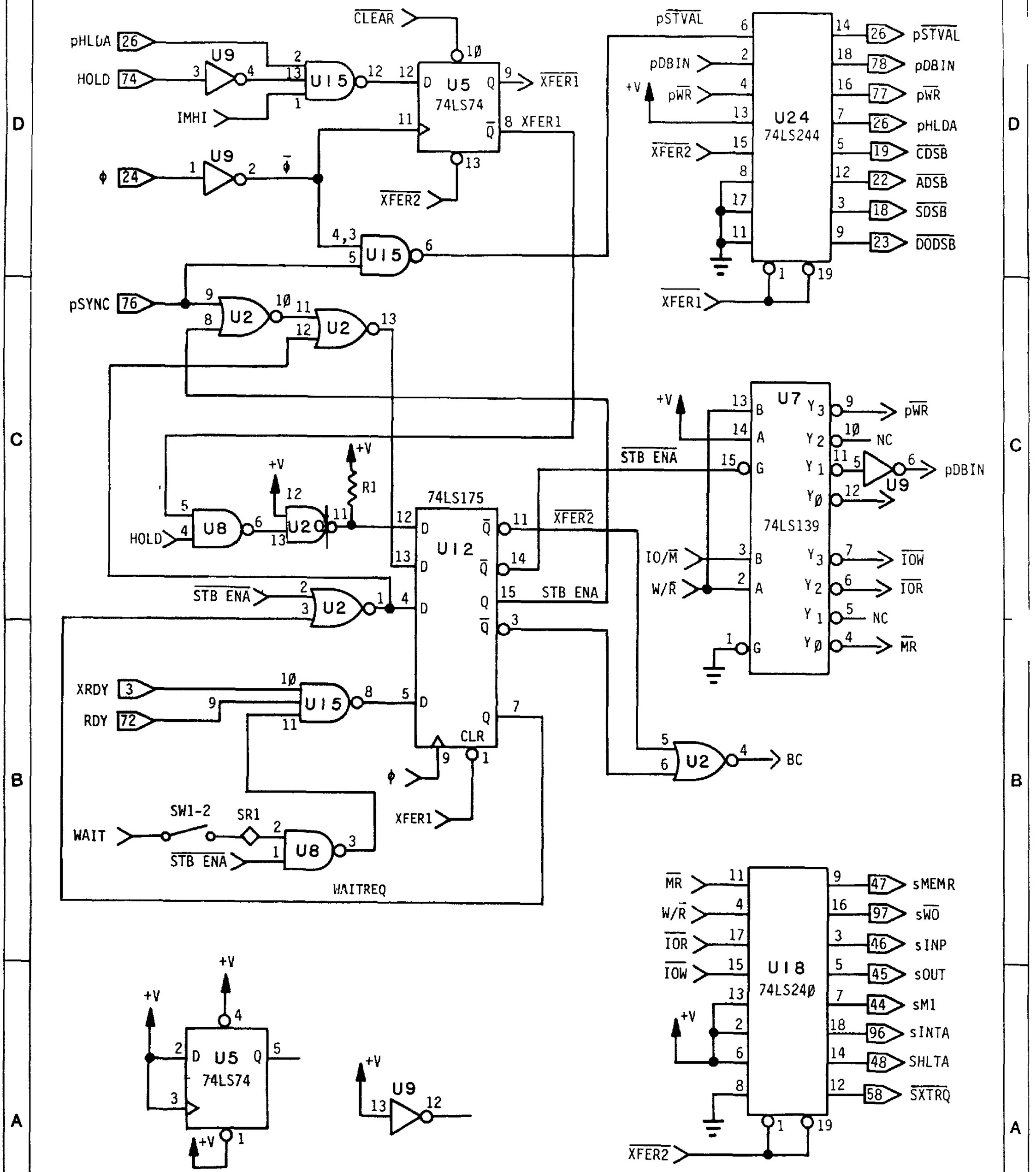
172D **©1981**

◇ = SIP PULL-UP RESISTOR = Vcc / SR X



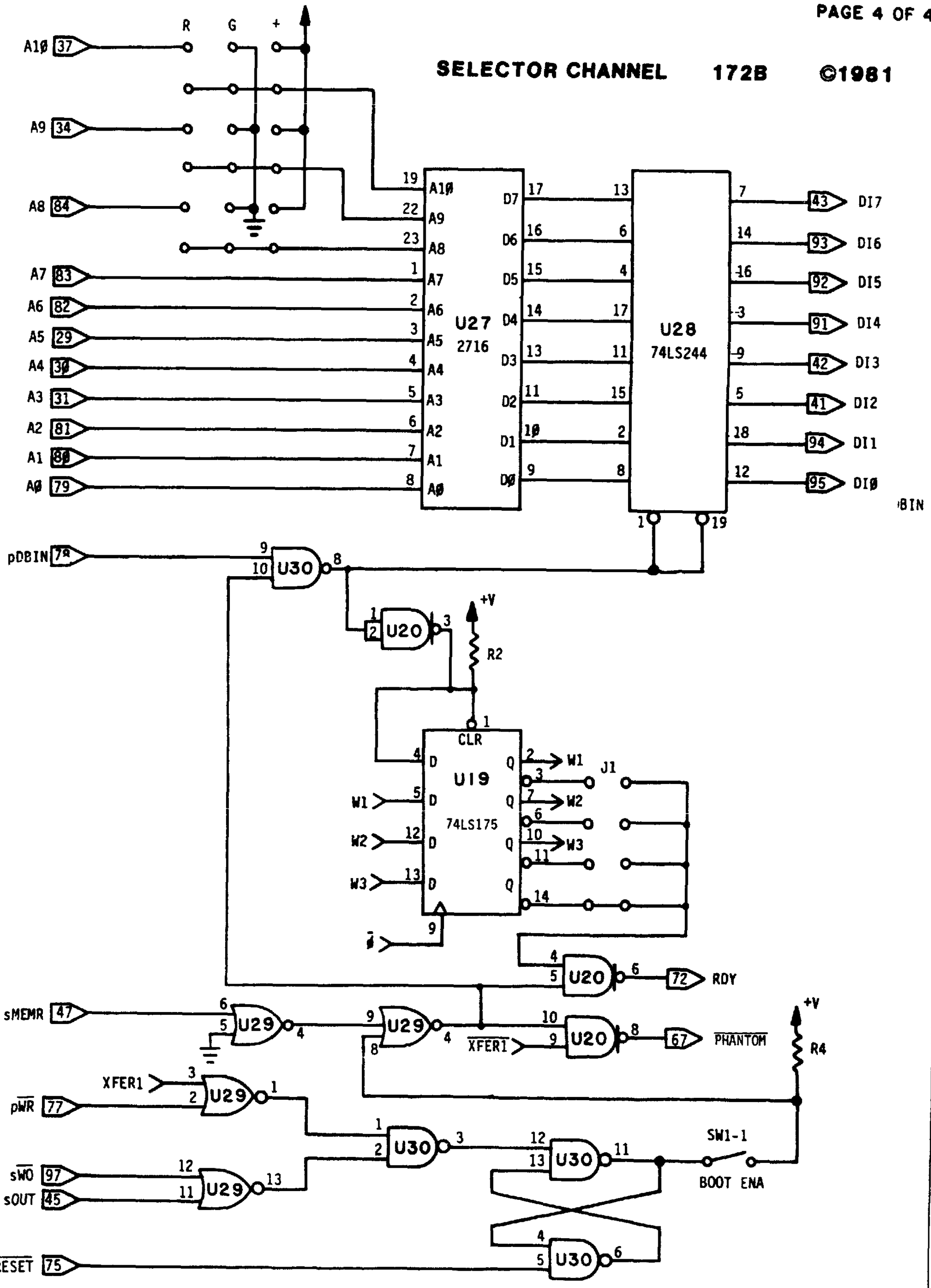


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PARTS LIST

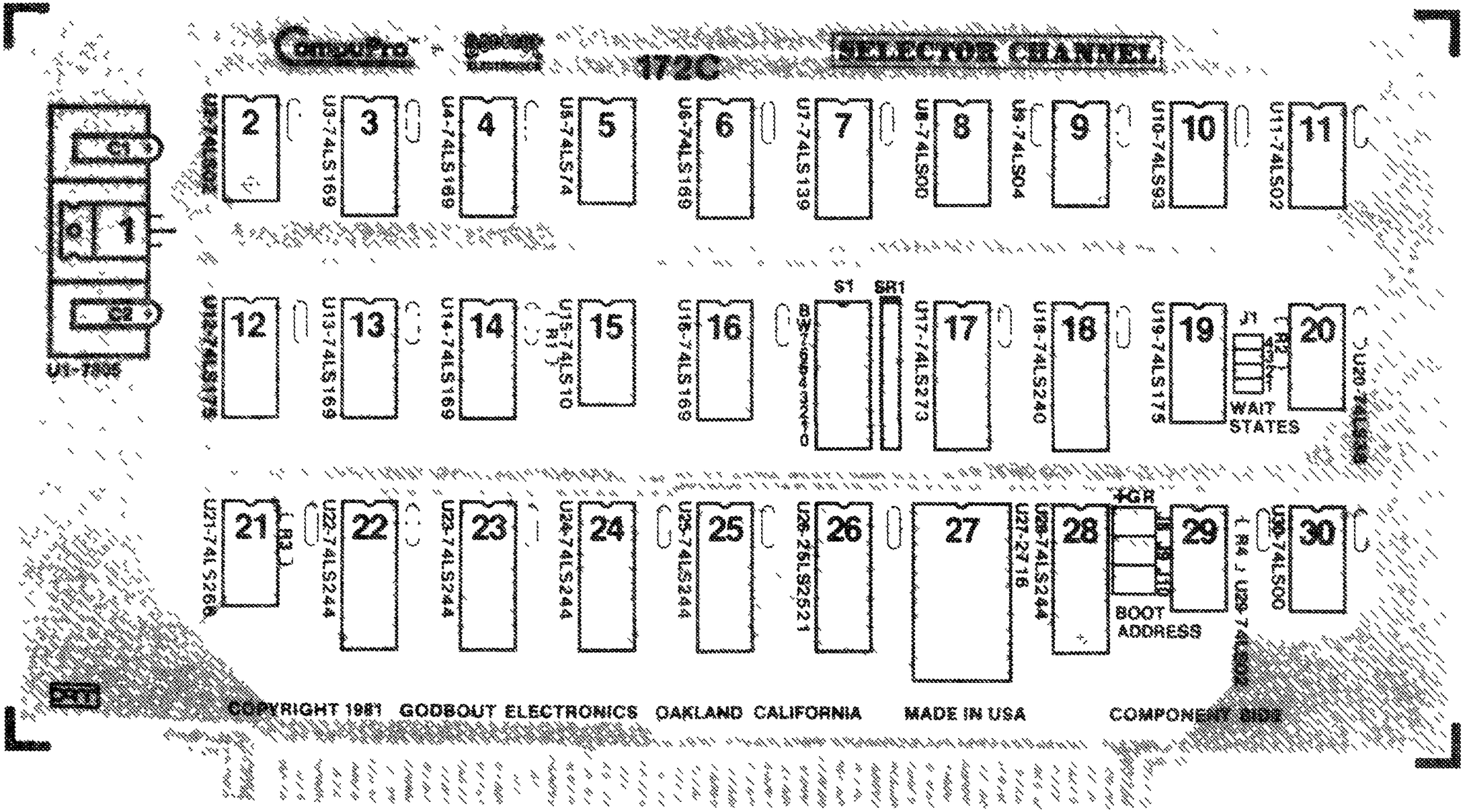
SEMICONDUCTORS

UNIT#	PART#
U1	7805
U2	74LS02
U3	74LS169
U4	74LS169
U5	74LS74
U6	74LS169
U7	74LS139
U8	74LS00
U9	74LS04
U10	74LS93
U11	74LS02
U12	74LS175
U13	74LS169
U14	74LS169
U15	74LS10
U16	74LS169
U17	74LS273
U18	74LS240
U19	74LS175
U20	74LS38
U21	74LS266
U22	74LS244
U23	74LS244
U24	74LS244
U25	74LS244
U26	25LS2521
U27	2716 (optional)
U28	74LS244
U29	74LS02
U30	74LS00

RESISTORS

	VALUE
R1	4.7K Ohm
R2	4.7K Ohm
R3	4.7K Ohm
SR1	5.1K Ohm

NO COMPONENT CHANGES BETWEEN REVISIONS C AND D



COMPONENT LAYOUT

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